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“Pal-Pazzar“

Mobile application for Palestinian Products

متجر لبيع المنتجات الفلسطينية

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ملخص الدراسة

فكرة تطبيق بال بازار ريادية تتمحور حول المنتجات الفلسطينية والحفاظ عليها، لقد قمنا بتطوير تطبيق متجر الكتروني يعرض المنتجات الفلسطينية التي تقدمها الشركات الناشئة (خشبيات، تطريز، مشغولات يدوية، هودي، إلخ) في مكان واحد بهدف أساسي وهو الحفاظ على التراث الفلسطيني وهويته مع تسويق المنتجات وتسهيل وصول العملاء لها، إن ما يميز تطبيقنا أنه يحمل هدفًا قيمًا عزيزًا علينا،

ألا وهو "فلسطين معنا في كل شيء"، حيث نعتز بكل شيء يدل على فلسطين من حولنا، وفي تطبيقنا العديد من الميزات وهي: منتجات يدوية فلسطينية في مكان واحد، سهولة الوصول للمنتج المطلوب من خلال البحث عنه، تحديد أكثر من عنوان لإرسال الهدايا للأصدقاء عبر التطبيق، يدعم عدة لغات، يدعم السمة الداكنة والفاتحة. يساعد نوعين من المستخدمين: مزودو الخدمات، والعملاء، حيث يمكن لمزود الخدمة إضافة المنتجات بكل سهولة من خلال لوحة التحكم، وإدارة عملياته بأريحية تامة، والعميل يمكنه الشراء من خلال التطبيق والعثور على المنتجات التي يريدّها. تم اتباع منهجية البروتوتايب في التطبيق، عملنا وفق خطة واضحة في تحليل التطبيق ودراسة سوق العمل وعمل مقابلات ومقارنات عديدة، وصولاً إلى التصميم والتنفيذ والبرمجة ومن ثم الاختبار.

Abstract

The idea of **Pal-Pazzar** application is a pioneer centered on Palestinian products and their preservation, we have developed an online store application that displays the Palestinian products offered by startups companies (wooden, embroidery, handmade, hoodies) in one place with a primary goal of preserving Palestinian heritage and identity with marketing products and facilitating customer access to them. That distinguishes our application is that it carries a valuable goal dear to us, which is *“Palestine is with us in everything”*, where we cherish everything that indicates a Palestinian around us, and in our application, there are many features such as handmade Palestinian products in one place, easy access to the required product by searching for it, specifying more than one address to send gifts to friends through the application, supports multiple languages, supports dark and light theme. It helps two types of users: business owners, and customers, where the business owner can add products easily through the dashboard, and manage its operations with complete comfort, and the customer can buy through the application and find the products he wants. The prototype methodology has been followed in the application, we worked according to a clear plan in analyzing the application, we did the market study, conducting numerous interviews and comparisons, and leading to the design, implementation, programming, and then testing.

إهداء

إلى والِدَيِ الغالِبِينَ اللّذين لولاهُما لَمّا أبصرتُ الطّريقَ، قوّتي في هذه الحِياة وسَندي ومصدر إلهامي..

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أخواتي وإخواني من كنتُ ولازالَتُ أستمُدُّ منهم عزيمتي وإصراري.

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فالحمدُ لله كلّ الحمد أن بلّغتنا يا رب هذا اليوم..

اليومُ المنتظر من أربع سنين..

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List of Abbreviations:

API	Application Programming Interface
IDE	Integrated Development Environment
IA	Information Architecture
UX	User Experience
ERD	Entity Relationship (ER) Diagram
UI	User Interface
FAQ	Frequently Asked Questions
CVC	Card Verification Code
GPU	Graphics Process Unit
OS	Operating System

Chapter 1

Introduction

1.1 Introduction

E-services are a highly generic term and it's the most trend in our daily lives, e-services constitute the online services available on the internet.

Due to the high popularity and fast expansion of the Internet and network technology, e-commerce has become an essential activity in contemporary business operations. Today, massive business activities are conducted online. People go online to sell and buy both goods and services, and many transactions cannot be completed without Internet technology [1]. According to this, people have taken the idea of online shopping so that it has become a natural thing and part of the shopping and buying habits of the entire world, which has undoubtedly become an imposing consequence, that e-commerce has been growing fast and non-existent since the 1990s. Because they have been in their simplest form ever since and have not stopped growing because of their importance and the growing demand for them, that enables us to say very easily that online stores are becoming increasingly important and are becoming an integral part of the global market.

Because Gaza has become buzzing with startup companies that are sold Palestinian products for the last five years, this has affected the behavior of users and their lack of knowledge about most of these companies because they are so numerous, on the other hand, their inability to find Palestinian products in one place, also, the user's inability to find what he is looking for at the right time, and thus taking too much time when purchasing.

To help everyone in the Gaza Strip, from users who suffer from the availability of all Palestinian products in one place and difficulty of using the browser on the mobile in case the startup owner has a website, we have developed an application and dashboard named "Pal-Pazzar" to manage the operations between customer and business owner.

Using Pal-Pazzar application, you can reach as many more categories as you can with ease and less effort. In addition, you can find what you're looking for and access it on time and complete your task faster.

Pal-Pazzar helps you to shop and know all the available Palestinian products and trends in the market.

Furthermore, we have proposed another application named “ Pal-Pazzar Delivery” in order to help the business owner to know and confirm the arrival of the order to the customer by the third-party delivery service by sending an SMS message to the business owner with the order number that was successfully delivered to the customer.

To sum up, our application will not only help you get quick and easy access to Palestinian products, but it will also help you to have an excellent user experience in browsing the products.

1.2 Problem Statement

Due to the tremendous and rapid development in the world of technology and computerized systems and the increase in awareness of the importance of technologies around us, the use of mobile applications by companies and organizations has become essential to keep pace with the development and increase the company's sales. With the advancement of technology, e-commerce applications that provide an easy way to sell products to a large customer base, and store owners and their products have had a share of this development due to the increasing problems that users face when purchasing a particular product. such as the difficulty of promoting and marketing the products that small business owners, also to preserve Palestinian products identity, and collecting them in one place, in addition to the problem of the user's navigation between the pages he needs and the difficulty of identifying all his purchases. As a result, Pal-Pazzar application aims to overcome for mentioned challenges.

1.3 Project Objectives

- Enhance the custom experience in terms of ecommerce by proposing new application named 'Pal-Pazzar' to facilitate the ordering and browsing process for the customer.
- Helps the Palestinian product provider by start-ups to promoting and marketing their products from one place.
- Enhancing the business owner experience in terms of delivery service by proposing a new application named "Pal-Pazzar Delivery", for tracking purposes.

1.4 Project Features

- **Accessibility:** It helps you to find whatever is Palestinian in one place such as hoodies, embroidery, wooden and handmade.
- **Searching:** Using the search feature, you don't have to worry about finding the product you're looking for within seconds.
- **Adding new address:** Pal-Pazzar application not only helps you choose and buy the products you want but also enables you to buy and send gifts to friends with the ability to specify the address to which the order is to be delivered.
- **Different kinds of payment methods:** Using Pal-Pazzar app, you can choose the appropriate payment method for you, either through an e-payment using Visa or cash on delivery.
- **Supporting multiple languages:** Pal-Pazzar application supports both Arabic and English languages.
- **Supporting the dark and light theme feature** that many users prefer because it is reducing pressure on the eye.
- **Provide a special application named "Pal-Pazzar Delivery"** for third-party delivery service company and business owner.

1.5 Project Significance

Our application will assist various types of customers in many ways. Here, we discuss the importance of our application to business owners, customers, and third-party delivery service.

- **Business Owner:** The provider can easily add its products through the dashboard and the ability to easily organize and manage its operations and

respond quickly to customers, as well as its ability to promote its products more widely both locally and globally.

- **Customers:** The customer can use this app to find all Palestinian products easily and thereby save the time and effort it takes to navigate between pages and reach what he is looking for in a timely manner and accomplish his task faster. In addition, preserve Palestinian identity by gathering many Palestinian products in one place.
- **Third-Party Delivery Service Company:** The business owner can know and confirm the arrival of the order to the customer by the third-party delivery service by sending an SMS message to the business owner with the order number that was successfully delivered to the customer.

1.6 Project Scope

Figure 1.1 shows the project scope for Pal-Pazzar app.

The geographic location scope is Gaza, which combines Palestinian product sellers business owners with customers looking for these products.

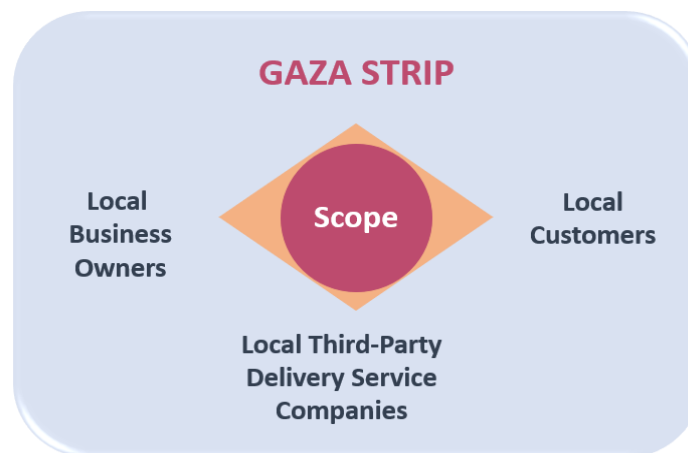


Figure 1.1 Project

1.7 Methodology

In our application, during its life cycle, we used the prototype methodology, which is one of the common methodologies used in software development, creates a basis for the production of the final program, where the prototype of the program is built and then tested if it meets the requirements or not, and then reformulated until we reach a good and acceptable prototype.

This methodology works best in scenarios where project requirements are unknown and predefined. Assists in understanding and defining customer requirements at an early stage of product development. It helps software developers understand what is expected of the product under development and obtain customer opinions and feedback about the product and its performance. It helps software designers and developers understand exactly what is expected of the product under development. It is mainly a trial-and-error process that operates in an iterative manner between the client and the software developer.

The prototype Model has 6 phases, which are shown in the Figure 1.2:

- 1. Requirement Aanalysis.**

In this phase, we identify the customer needs and evaluate the project feasibility after conducting group interviews.

- 2. Design.**

In this phase, we transform all requirements and features into a complete wireframe design, then design the user interfaces based on UX principles [7].

- 3. Building and Prototyping.**

Build the first prototype for the application based on the information gathered.

- 4. Evaluation by Customer.**

In this phase, we present the prototype for the user to recognize the strength and weakness, and what we can edit or remove.

5. **Refining Prototype.**

In this phase, after the user evaluate the current prototype, we develop a new prototype with the additional information added by the customer, this process continues until all the requirements specified by the customer are met.

6. **Final Product.**

In this phase, the final product is developed based on the final prototype, when the customer was satisfied with the final prototype.

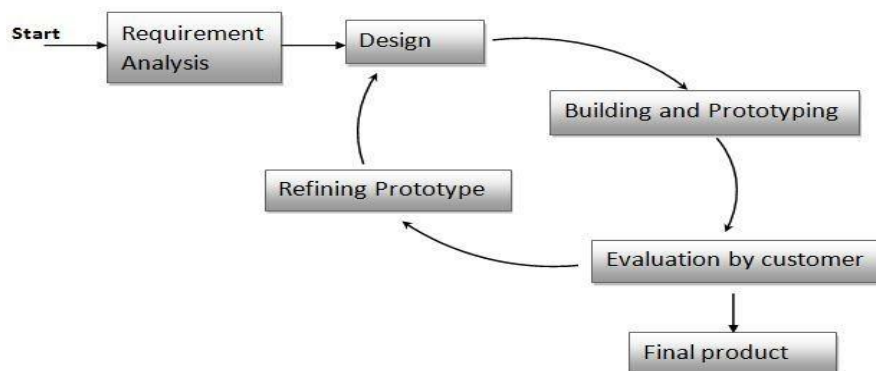


Figure 1.2 Prototyping Model Phase.

1.8 Time Table

Figure 1.3 shows the time table of Pal-Pazzar App. It specifies the start time, end time and duration for each phase.

Title	Type	Duration	Start	End	%	Assigned to	+ v
Documentation	T	220 days	01/10/2021	04/08/2022	0%		
Requirements Gathering	T	9 days	23/09/2021	05/10/2021	0%		
Project Planning	T	7 days	07/10/2021	15/10/2021	0%		
Analysis	T	24 days	12/10/2021	12/11/2021	0%		
Design	T	42 days	01/01/2022	01/03/2022	0%		
Implementation	T	78 days	03/04/2022	20/07/2022	0%		
Testing	T	10 days	22/07/2022	04/08/2022	0%		

Figure 1.3 Time Table

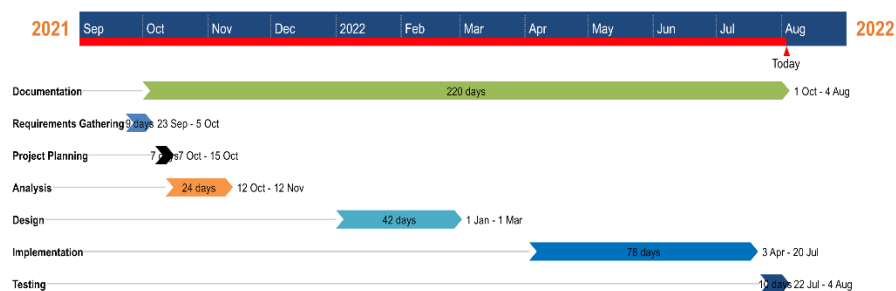


Figure 1.4 The Gantt Chart for the Time Table.

Chapter 2

Literature Review

2.1 Introduction

The need for the organization and operation of the project and various resources appeared, and this science has developed to this day; Therefore, the agencies have conducted many studies and research to find suitable solutions to address these issues and help people in facilitating the online application process in a timely manner , the term e-commerce has emerged and means that all online activities that involve buying and selling products and services provide quick delivery of goods with very little effort on the part of the customer.

About 21.8% of the world's population shop online, the percentage of establishments that conducted commercial transactions by electronic means reached 11.2% of the total establishments in the Palestinian territory in 2011; while the percentage of establishments that conducted electronic commercial transactions was 4.2% of the total establishments in 2009 compared to 2.3% in 2007. While in 2019, there were about 900,000 parcels from abroad, 85% of them came from China via Palestinian Post [2].

The growth rate between 2018 and 2019 was about 40%. This market has become an important growth factor, as it is a promising space, especially in light of the trend towards the digital world. E-commerce sales are expected to reach 17.5% of retail sales by 2021, According to Forrester, B2B e-commerce in the US will reach \$1.8 trillion by 2023[2].

There are many e-commerce applications open source in platform such as iBazaar, Bazaar.om, Vendr, Platform offers.

In the upcoming section, for mentioned related apps are presented and discuss.

2.2 Related Works

2.2.1 iBazaar:

As shown in Figure 2.1, it is an application that displays all commercial and marketing activities within the state of Iraq, as it allows them to facilitate the sale and purchase of all types of goods. The application supports several languages, the most important of which are (Arabic, English and Kurdish), it has the feature of adding an advertisement to the project owner for free, and it has a search box for opportunities Work, the application includes some cities of Iraq and not all cities, unlike our application that includes all the governorates of the Gaza Strip [3].

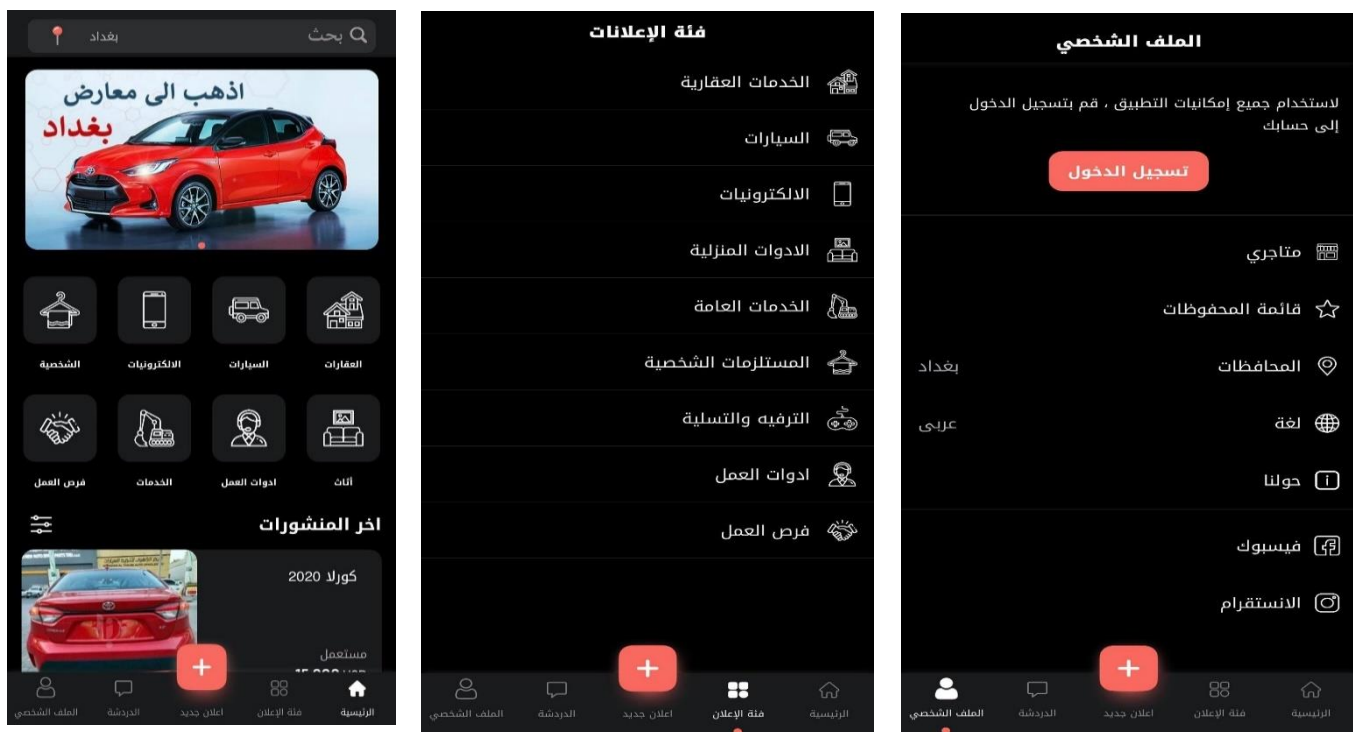


Figure 2.1 iBazaar Application Interfaces

2.2.2 Bazaar.om

As shown in Figure 2.2, Bazaar.om in Oman, the online supermarket segment with grocery and fresh food. With over thousand products in our category, you can be rest assured to have everything that you are looking for.

Bazaar.om is an Online Start up Backed by “Tamimah Group” which is an entrepreneurial company established in 1984.

The company Bazaar.om, the E-commerce arm of “Tamimah Telecom” runs on a fast and lean supply chain – from producers/Wholesalers straight to end customers. Right from Mobile and accessories, fresh fruits and vegetables, dairy products, spice and seasoning, baby products, confectionaries, juices, personal care products, fresh meat and chicken [4].

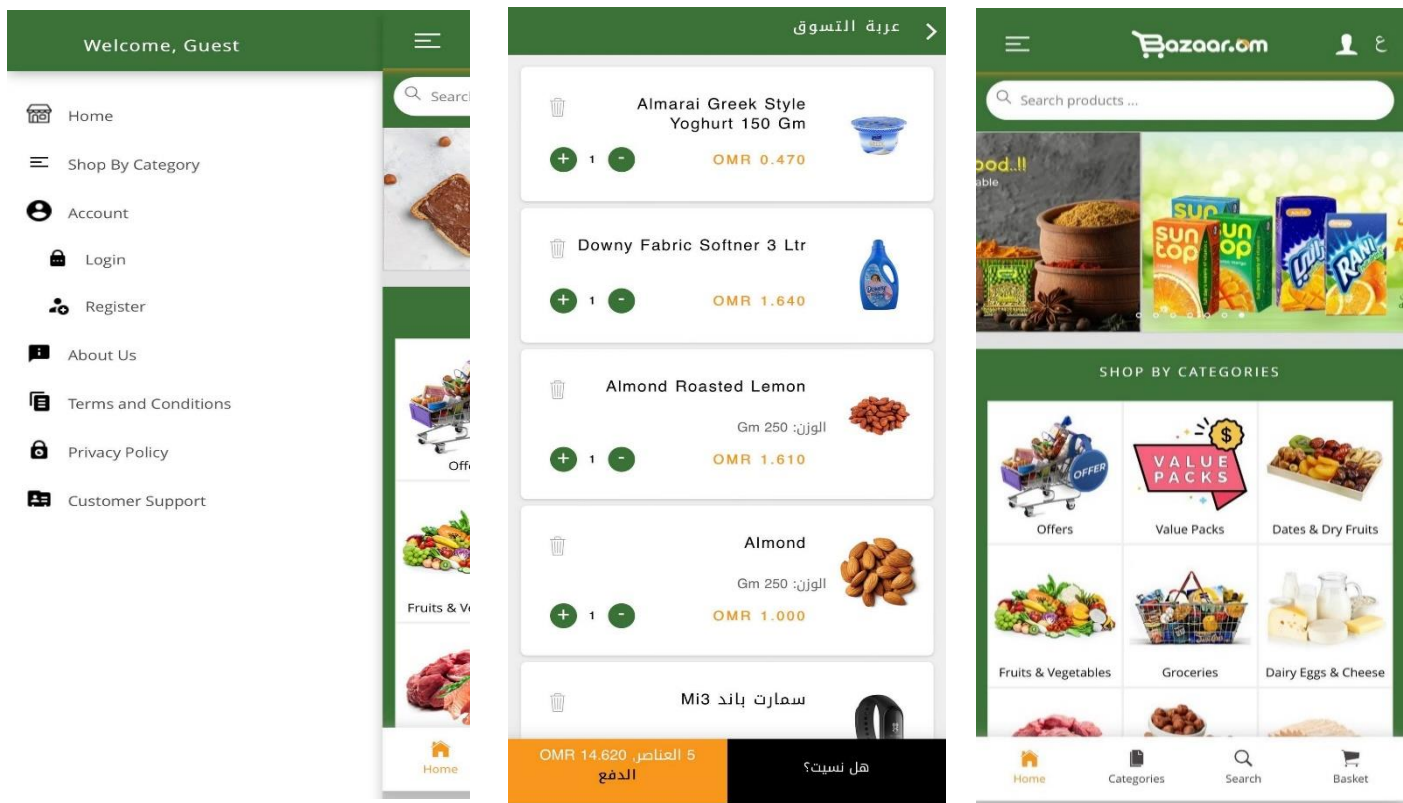


Figure 2.2 Bazzar.om Application Interfaces

2.2.3 Vendr:

As shown in Figure 2.3, It is classified as the first e-commerce application in the Gaza Strip As an initial site in the Corona crisis, it was easy to use, but it was not developed, as it is shortened to a specific group of restaurants and pharmacies and only one supermarket (metro), unlike our application that combines more than one category through which you can shop comfortably and easy to use [5].

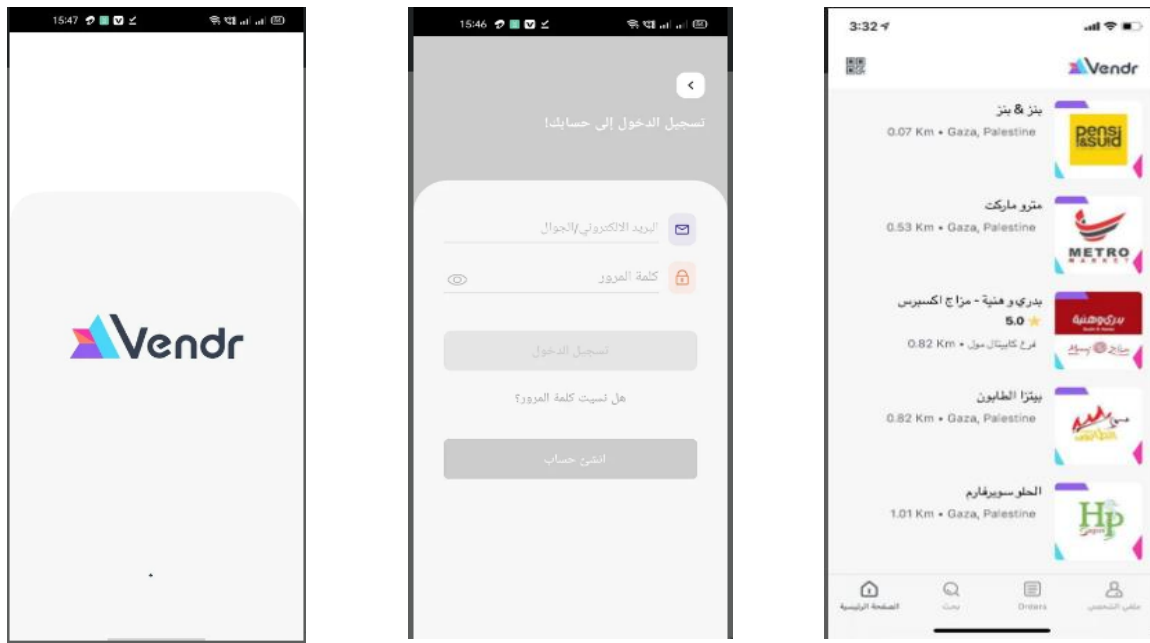


Figure 2.3 Vendor Application Interfaces

2.2.4 Platform offers:

As shown in Figure 2.4, It is an application in the Gaza Strip for online shopping provide many categories such as (restaurants, clothes, perfumes), but with very limited and few stores, as it allows you to arrange your list of purchases while specifying the store from which you purchased, but it does not support Palestinian products, unlike our specialized application Palestinian products. Currently, the application is not working, and there is no known reason yet why it stopped [6].

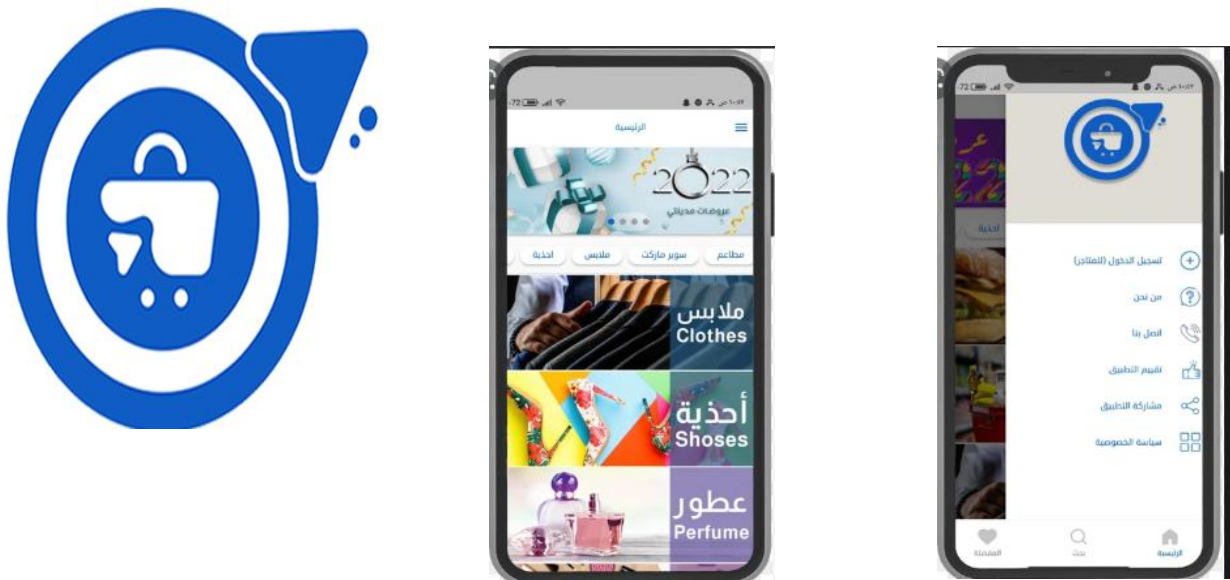


Figure 2.4 Platform Offers Application Interfaces

2.3 Comparison between the State-of-the-Art Applications:

In this table, we have compared the best applications and put some features that distinguish us from others, such as (multiple payment methods, search feature, Ability to add new address, having a FAQ option, support multiple language, and provide ability to deliver the purchase to many provinces).

Table 2.1 Comparison Between the State-of-the-Art Applications

	Pal-pazzar	Bazaar.om	iBazzar	Vendr	Platform offers
Multiple payment methods	YES	NO	YES	NO	NO
Search feature	YES	YES	YES	YES	YES
Ability to add new address	YES	NO	NO	YES	NO
Favorite to Add	YES	NO	NO	YES	YES
Having a FAQ option	YES	NO	NO	NO	NO
Support multiple language	YES	YES	YES	NO	NO
Provide the ability to deliver the purchased to many provinces	YES	NO	NO	YES	YES

Chapter 3

Methodology

3.1 Introduction

In this chapter, we discuss the requirements, analyzing them starting with the determining the target sample also preparing the interview questions then conducting the interviews.

We did requirements analysis from customers & business owners, and we specify non-functional requirements: performance, data integrity, security, usability requirements, compatibility, and maintainability.

In addition, user stories from customers & business owners, was discussed scenarios of customers & business owner stories in the application to describe the processes and what they do in the application in each situation & descriptions of unexpected situations for each process to make a correct decision for it to achieve the goal of the App.

3.2 Requirements Gathering and Analysis

In this section, the requirements of the App are defined in detail as follows:

Step 1. Determining the target sample:

During this stage, a random sample of those interested in buying and selling online was identified.

We prepared a questionnaire and interview sample of customers and business owner to check their needs and achieve their aspirations from the App.

Step 2. Preparing the interview questions:

In this stage, we prepared questions that we want to ask to determine their needs.

Table 3.1 shows the questions of the questionnaires.

Table 3.1 Questions of The Questionnaires.

No.	Customers	Business Owners
1.	Do you prefer buying from the market or buying online?	Do you prefer selling your products in the market or online?
2.	What drives you to buy online?	What drives you to sell online?
3.	What are the most products you buy online?	What kinds of products that you sell online?
4.	Do you like to buy handmade products (clothes, and gifts)? and why?	What distinguishes your products from others and encourages people to buy them?
5.	What do you think when you see the Palestinian handmade products were exported abroad?	What do you think when you see the Palestinian handmade products were exported abroad?
6.	What features and services would you like the App. to provide for you?	What features and services would you like to have in the App.?
7.	What problem do you face when buying a Palestinian product (handmade, embroidered and wooden) online?	What are the drawbacks and obstacles that you will face when selling online?

Step 3. Conducting the interviews

After preparing the questionnaire, we conducted many interviews with the target sample of customers and business owners.

We have collected and selected the most popular answers and then analyzed them, as shown in Table 3.2.

Example: Question 1: "As a customer, do you prefer buying from the market or buying online?" 15 customers answered: marketplace and 10 customers answered: online.

Another example: Question 6: "As a business owner, what are your products that you sell online?".

5 business owners answered: food,

7 business owners answered: handmade,

10 business owners answered: Clothes

And 5 business owners answered: Skin care products.

Table 3.2 The Most Common Answers of Interviewees.

No.	Question	Answer
1.	As a customer, Do you prefer buying from the market or buying online?	From the market.
2.	As a business owner, do you prefer selling your products in the market or online?	Online.
3.	As a customer, what drives you to buy online?	- Saving time and efforts. - Get special and unique products from anywhere.
4.	As a business owner, what drives you to sell online?	- Easier to sell. - Lower costs in terms of space and rent, as products are displayed and sold online at reasonable costs.

		- Selling products and increasing the demand for them.
5.	As a customer, what are the most products you buy online?	- Clothes. - Food. - Gifts.
6.	As a business owner, what kinds of your products that you sell online?	- Food. - Handmade. - Candy. - Clothes. - Pastries. - Gifts. - Candles. - Skin care products.
7.	As a customer, do you like to buy handmade products (clothes and gifts)? and why?	Yes, because most of these products are distinctive, made with precision and beauty of detail, and are often not available in non-handmade products.
8.	As a business owner, what distinguishes your products from others and encourages people to buy them?	Credibility in products, with the best quality and at a competitive price in the market.
9.	What do you think when you see the Palestinian handmade products were exported abroad?	Being Palestinian, we feel proud and happy about it, because we represent our Identity.
10.	As a business owner, what features and services would you like to have in the App.?	- Facilitate the ordering process and reach customers in the shortest possible time.
11.	As a customer, what features and services would you like the App. to provide for you?	- The app should be simple and easy to handle. - Fast delivery. - Reliable products. - Full product description.
12.	As a customer, what problem do you face when buying a Palestinian product (handmade, embroidered and wooden) online?	- The order was late in arriving. - The order does not match the expected product image.

		- Long delivery time. - low product quality.
13.	As a business owner, what are the drawbacks and obstacles that you will face when selling online?	- The customer's lack of seriousness after order.

Step 4: Requirements Analysis

After preparing the questionnaires that were filled out by the customers and business owners, we reached the results and based on them determine the outputs customers and business owner needs.

We present the statistics based on the analysis of the questionnaire, as shown in Figure 3.1, Shows the percentage of people who prefer to buy from the market or buy online and Figure 3.2 , Shows the reasons why people motivate and encourage them to buy online.

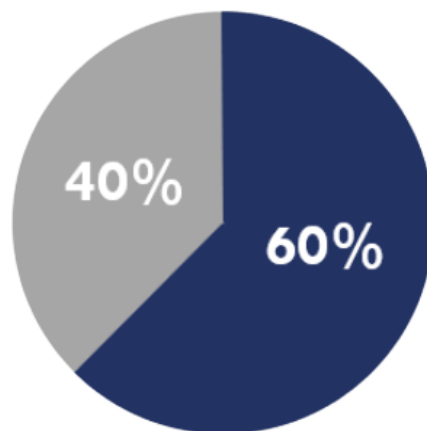


Figure 2.1 Question No.1

Do you prefer buying from the market or buying online?

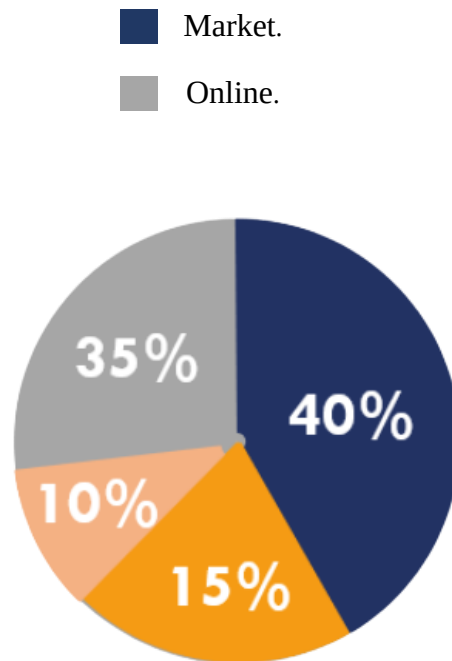
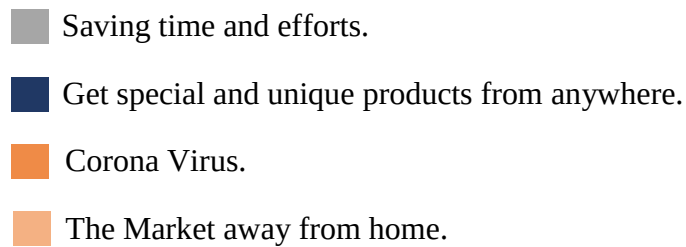


Figure 3.2 Question No.3

What drives you to buy online?



3.3 Functional Requirements from User:

- The users sign up for the App by name, phone number, password, gender, and city.
- The user verifies their account by receiving a verification code from API.
- The user login by their phone number and passwords to start using the App.

- The user is able to browse all products with different categories and subcategories.
- The user can add products to their favorite.
- The user can add products to their cart.
- The user can change the language.
- The user can update their profile.
- The user can change their password.
- The user can send a message to the business owner.
- The user can see all orders and be able to increase or decrease the count.
- The user can see recent and adaptive FAQ and about app interfaces.
- The user can logout.

3.4 Functional Requirements from Business Owner:

- Can add new product for the users.
- Can add the number of categories he needs.
- Can update or delete previous products.
- Can see his incomes.
- Can see all products that he posted.
- Can see all contact requests from users.
- Can see all user's status is their verified or not.
- Can activate or deactivate the visibility of the product.

3.5 Non-Functional Requirements:

A non-functional requirement is a requirement that specifies criteria that can be used to check the operation of an App, rather than specific behaviors [8].

Pal-Pazzar fulfill the following properties:

Performance requirements

- The application load time not more than one second for users.
- The App has high performance that can control the number of users without delay, cancellation, or suspension.

Data Integrity

- The App maintain data integrity by keeping backups for all updates to the database for every record transaction.
- The App not change by unauthorized users, and the App maintain trustworthiness, consistency, and accurate information by check if the user verify their account or not and give the permissions.

Security requirements

- Account Creation: App may require users to create accounts to access applications that store information and display profiles. A security system typically grants access to accounts when users enter the correct phone number and password.
- Password generation: An application may not grant access until the user creates a strong password, Pal-Pazzar password should contain at least five digits (numbers, characters, and symbols).

Maintainability requirements:

- Pal-Pazzar application guarantees good performance and can work properly under traffic by deploying on high storage hosting.

Usability requirements:

A design usability depends on how well its features accommodate user needs and contexts, it should contain these elements [9]:

- **Effectiveness:** Pal-Pazzar supports this requirements by make users completing actions accurately.
- **Error Tolerance:** Pal-Pazzar support this requirements by show error and success messages.
- **Ease of learning:** Pal-Pazzar support this requirements for new users by providing on-boarding and about app interfaces.

Compatibility:

- Pal-Pazzar allows people who have different operating systems to use the same application.

3.6 User Stories

A user story is an informal, general explanation of a software feature written from the perspective of the end user. Its purpose is to articulate how a software feature will provide value to the customer [10].

3.6.1 Customer User Stories

- As an unregistered user, I want to sign up so I can use full application features.

- As a registered user, I want to login so I can use full application features.
- As a registered user, I want to restore my password in case I forget it.
- As a registered user, I want to update my profile so I can keep my profile up to date depends on my status.
- As a registered user, I want to see all my purchases.
- As a registered user, I would like to view all categories and subcategories.
- As a registered user, I want to logout of the account.
- As a registered user, I want to select the appropriate payment methods.
- As a registered user, I want to add some products to my cart and favorite.
- As a registered user, I want to change the language of the App.
- As a registered user, I want to contact with provider (Pal-Pazzar owner).
- As a registered user, I want to read the FAQ , so that will help me use app easily.

3.6.2 Business Owner User Stories

- I want to add a new product for users.
- I want to add categories and subcategories.
- I want to see all my user's status (verified or not).
- I want to change the visibility of my products.
- I want to update the details of a previously added products.
- I want to delete a previously added product.
- I want to know the number of products, customers and inquiries.
- I would like to know all information about order.
- I can see total incomes.

3.7 Scenarios

In the following section, we discuss scenarios for user stories, which is a detailed description of what users do with the product and more importantly why they do it, and it is a real-world narrative of the process to accomplish a user's goal. A user scenario will articulate the goal, the process, and the outcome that is contextually relevant to the user [11].

3.7.1 Create a New Account

When the user clicks on the register button, the registration form pops up, which displays the data required for the registration process (name, phone number, password, city, and gender), the user sends his registration information, then the membership activation code is shown with a message that appears inside the application, if the activation code and user input are verified correctly, the user will be registered in the system and added to the database.

- Unexpected situations:

- Incorrect input data.
- Timeout of activation code.

3.7.2 Sign in using Phone Number and Password

If the user registered his data before that and wanted to log in again, then he is asked for his phone number and password, and after that, he is able to use the App.

- Unexpected situations:

- Phone number or password may be entered incorrectly.

- Forget the password.

3.7.3 Contact with Business Owner:

When the user needs to send feedback or needs to send any message to the business owner, he can do that by clicking on the contact us icon.

- **Unexpected situations:**

- The user is unable to send a message to the business owner because the connection is lost.

3.7.4 Order Products

When the user clicks on the checkout button on the cart page, he will navigate to the add new address page, which displays the data required for this process (name, phone number, address, and city), then he clicks on the add new address button for the first time, which will navigate him to choosing payment method page (Visa or cash on delivery). If he chooses the visa, he will navigate to the add new card page, which displays the data required for this process (card number, CVC, and validity date), then he clicks on the add card button. Finally, an order summary form pops up, which displays all the data previously entered (products, address, payment method, and total price), when the user clicks the confirm order button, the application sends the order to the business owner.

- **Unexpected situations:**

- User's inability to complete the order because he isn't adding his payment method and his address.
- Input data does not match the requirements or some field is left empty.

3.7.5 Add to Cart

If the user wants to order more than one product at the same time before completing the purchase process, he can click on the add to cart button on the product details page, which enables him to add all his products to the cart easily instead of ordering each product separately and thus consuming a lot of time and effort. If he clicks on the add to cart button on the product detail page, the cart page will pop up, which displays all the products that he wants to buy, with the possibility of deleting any product he no longer wants or adding this product to the favorites list to buy it when needed.

3.7.6 Update Profile Information

There is an icon to update the profile and it can change information that he previously entered and wants to update, such as name, phone number, email, and password. When he finishes updating the data, he clicks on the save icon, so his new data is saved.

- Unexpected situations:

- It is possible that the data is not updated due to a field that does not match the requirements or that the field is left empty.

3.7.7 Add New Product

When the business owner clicks on add new product button, it will move him to a new screen that includes detail about the product like name, image, description, and price.

- Unexpected situations:

- The business owner doesn't fill all text fields on the screen, all text fields are required to save the product on the database.

3.7.8 Orders Option

When the business owner clicks on orders, it will move him to the orders interface and display all orders from users.

- **Unexpected situations:**

- There is no order at this time, so it will be an empty screen.

3.7.9 Total Incomes

In the main interface of the dashboard, it will display the income of the store owner.

- **Unexpected situations:**

- The business owner is new and doesn't have any orders so the system will display 0 in his income.

3.7.10 Reviews from Users

The business owner can see all reviews and feedback from users by clicking on contact requests.

- **Unexpected situations:**

- The business owner doesn't have any reviews from users.

3.7.11 Edit Profile

The business owner can edit his profile info and edit store info.

- Unexpected situations:

- It is possible that the data is not updated due to a field that does not match the requirements or that the field is left empty.

Chapter 4

Analysis and Risk Management

4.1 Introduction

In this chapter, we discussed analysis and risk management of Pal-Pazzar App.

The Pal-Pazzar App helps owners of simple projects in marketing their products.

We started by describing the App and explaining the value and features it offers to customers, and business owners and specified tools and App run requirements, then explained the use cases to customers and business owners, case by case, for example: sign up, login, add product. In addition, we have shown a sequence diagram were how to implement the processes in the App, and we have managed the risks in the App as it is a critical process for the success of the App and its plan and achieving the desired goals.

4.2 System Description

We offer the Pal-Pazzar App that helps its users to order their purchases with ease so that they can know what is being ordered. It also helps business owners to be more productive and creative and gives them good organization and management of their projects.

Also, in our App, we try to find a solution for simple projects owner by marketing their products using our App and providers all details about their sales and incomes every month also to make it easy for them to communicate with their customers and meet the customer needs.

4.3 Tools and App Run Requirements.

As shown in Table 4.1, illustrates the tools that we used to develop Pal-Pazaar App and Requirements that should be available to be run on different mobile phones OS, for example: we used Adobe XD to design the user interface in App, visual studio code to write the code, and postman for API management and testing.

Table 4.1: Tools and App Run Requirements.

Software tools	We used visual studio code (code editor), Adobe XD (UI design), and Postman (API management & testing).
Storage requirements	Storage for App data we used VPS hosting to store App data.
Memory requirements	We use asynchronous operations when retrieve data from server to be able to do another operation until complete the task, so that will make the App more efficient because there will not stop or crash and use the hardware in efficient way.
Input devices	Touch screen.
Output devices	Screen.
Security	application may not grant access until the user creates a strong password, and the account must verified before start using the App.
Network configuration	All the operation in the App need internet connection because we use API to retrieve data from the web server.

4.4 Use Cases

In this section, use cases for the entire App, which have been derived from the App requirements in section 3.3 are discussed in more details. In Figure 4.1 main use case is introduced firstly followed by each use case separately.

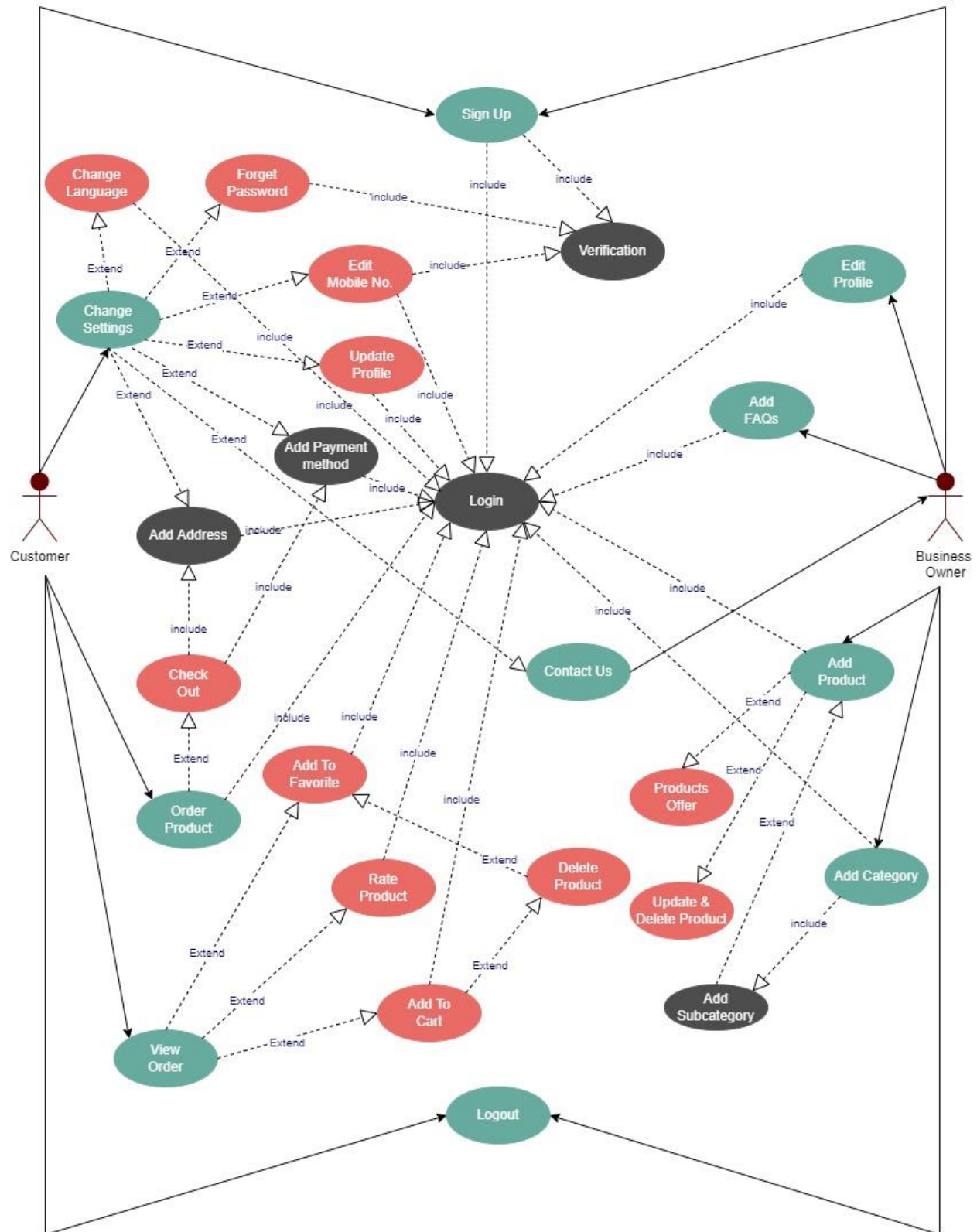


Figure 3.1App Use Case

4.4.1 Customer Use Case:

4.4.1.a Create a New Account Use Case (Signup)

Figure 4.2 shows create a new account use case which represents the first interaction between the customer and App.

The user creates a new account at the beginning to enjoy all the features of the application and order products, here the customer enters the name, city, gender, phone number, and then sends the verification number to the customer via App and enters it in the required field, and the application does not work without the Internet.

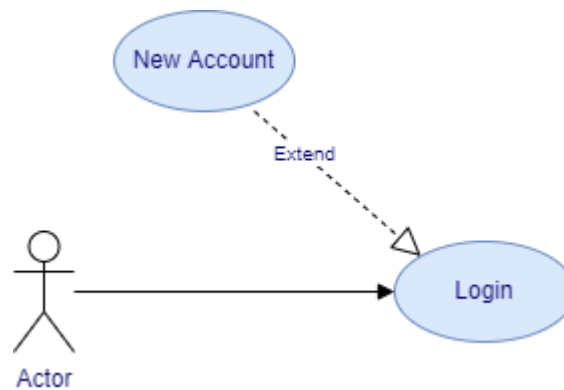


Figure 4.2 Sign Up Use Case

Table 4.2 Sign Up Use Case

ID	1
Name	Create a new account (Signup).
Description	The customer can create a new account in the App in order to be able to do the following action such as (order product, add product to cart, edit profile, add product to favorite).
Priority	High
Actors	Customers

Pre-conditions	-The device should be connected to the internet. -The user should have phone number. - The user should be verified before the registration process is completed.
Inputs	Customer name, phone number, password, gender and city.
Normal flow of events	-The customer enters his name ‘phone number‘ password‘ gender and city. -Home screen popups and user start browsing the products. - Customer profile is shown and user start editing his personal information when he clicks on the “Update Profile” button.
Otherwise, Flow	- The App. cannot connect to the server due to no internet connection or the user has entered wrong information. - The App displays an error message.
Post-Conditions	- The App. displays the home screen. - The App. displays the cart screen. - The App displays the user profile. - The App displays the favorites screen. - The App displays the payment screen. - The App displays the address screen. - The App displays the categories screen.
Outputs	None
Constraints	- Available internet connection. -Valid phone number .
Exceptions	- No internet connection. - Invalid Phone Number

4.4.1.b Login Use Case

The login screen is the screen that enables the customer to enter the application as shown in Figure 4.3 and through it, the customer enters the phone number and the password, through which he can make some requests such as adding to the cart or

favorites or updating the profile, if the customer is not connected to the Internet, the application does not respond.

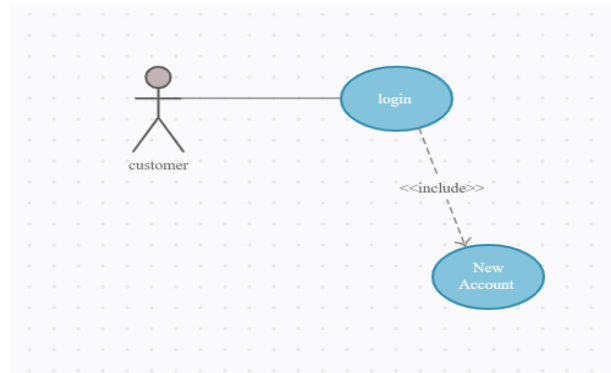


Figure 4.3 Login Use Case

Table 4.3 Login Use Case

ID	2
Name	Login
Description	The user can login in the App in order to be able to do the following action such as (order product, add product to cart, edit profile, add product to favorite).
Priority	High
Actors	customer
Pre-Conditions	- The device should be connected to the internet.
Inputs	User phone number and password.
Normal Flow of Events	-The user enters his phone number and password. - Main screen is shown and user start browsing the products

	when he used the app for the first time. - User profile is shown and user can editing his personal information, when he clicks on the “Update Profile” button.
Otherwise, Flow	-The app cannot connect to the server due to no internet connection or the user has entered wrong login information -The app displays an error message.
Post-Conditions	- The App displays the main screen. - The App displays the cart screen - The App displays the user profile. - The App displays the favorites screen. - The App displays the payment screen. - The App displays the address screen. - The App displays the categories screen.
Outputs	None
Constraints	- Available internet connection. -Valid login info.
Exceptions	- No internet connection. - Wrong login info.

4.4.1.c Order Product Use Case

A customer can order products from all categories, by choose the product he wants to order by adding it to a cart .Then interface to confirm the order and complete the order process by filling in the required data.

Table 4.4 Order Product Use Case

ID	3
Name	Order product.
Description	The user can order a product from different categories and subcategories.
Priority	High
Actors	Customer
Pre-Conditions	- User must be logged in first.
Inputs	Order completion form (name, phone number, payment method (cash or visa) and delivery address).
Normal Flow of Events	- The user chooses the products he wants to order by adding the product to the cart. - The user navigates to the cart to confirm the selection of products and complete the order. - The user navigates to the order completion form, where he enters the name, mobile number, delivery address, payment method (cash or visa) and clicks on the order confirmation button.
Otherwise, Flow	- Click on the complete order button on the cart page without choosing any product and showing a message “Choose a product first”. - Add invalid personal information in order completion form.
Post-Conditions	-When click on checkout button , a message appears that the order is complete. - The App displays the order in Orders screen.
Outputs	- New order is added in both sides (customer, business owner) list.

Constraints	- Successful login. - Valid personal info.
Exceptions	- Wrong login info. - Add wrong personal info.

4.4.1.d Add to Cart Use Case

The process of adding to the cart, when the customer adds more than one product to the cart during the browsing process before completing the purchase process, through the product details page, the product can be added to the cart easily and saves time and effort.

Table 4.5 Add to Cart Use Case

ID	4
Name	Add to cart.
Description	If the user wants to order more than one product at the same time before completing the purchase process, he can click on the add to cart button on the product details interface, which enables him to add all his products to the cart easily instead of ordering each product separately and avoiding consuming a lot of time and effort.
Priority	High
Actors	Customer
Pre-Conditions	- user must be logged in first.
Inputs	None
Normal Flow of Events	-The user clicks the add to cart button on the product detail page.

	- The product is added to the list of products on the cart page.
Otherwise, Flow	- The inability of the user to add its products to the cart because he did not log in first.
Post-Conditions	- The App displays products in the “Cart” screen. - The App increases the number of products on the cart icon. - After completing the order process, he can pay - The App displays the address screen and put the appropriate address for him. - The App displays the payment screen and selects the payment method. - A page appears containing the order summary such as (products, address, payment method) and then confirms the order. - A message will appear that the order has been successfully confirmed.
Outputs	- The product is added to the list of products on the cart page.
Constraints	-Successful login.
Exceptions	- Wrong login info.

4.4.1.e Add Delivery Address Use Case.

When completing the order process, the customer must put his own address for the first time and then he can also put more than one address and update or delete them.

Table 4.6 Add Delivery Address

ID	5
Name	Add delivery address.

Description	The user can add more than one address from setting screen and he can update or delete it.
Priority	High
Actors	Customer
Pre-Conditions	- User must be logged in first.
Inputs	Address name, info (brief description for the street and building name for example), contact number and city.
Normal Flow of Events	-The user enters the address name, info, contact number and city. - The address will be added in “addresses” screen.
Otherwise, Flow	- The inability of the user to add new address because he did not log in first. - Add invalid address information.
Post-Conditions	- The address will be added in “addresses” screen. - The address is added once at the beginning of using the application and thus makes it easier for the user to complete the purchase of the product without the need to add it again with each purchase.
Outputs	- The address will be added in “addresses” screen.
Constraints	- Successful login. - Valid address info.
Exceptions	- Wrong login info. - Invalid address info.

4.4.1.f Payment Method Use Case.

Upon completing the order process, the customer can select the payment method, as there are two options, either Visa or Cash, and when adding the Visa card to the application, it can be deleted or modified later.

Table 4.7 Payment Method Use Case

ID	6
Name	Payment method.
Description	The user can choose the payment method, either by Visa or cash when confirming the purchase, and can add a payment card details through the settings screen, in addition to the ability to modify and delete the card info .
Priority	High
Actors	Customer
Pre-Conditions	- User must be logged in first.
Inputs	Name, type (VISA), holder name, card number, expiration date and CCV.
Normal Flow of Events	- The customer enters the name, type (VISA), holder name, card number, expiration date and CCV. - The payment card details will added to the cards screen.
Otherwise, Flow	- The customer has entered wrong login info. - The customer has entered wrong card info.
Post-Conditions	- When choosing a payment method, two options appear, either paying on receipt or paying via Visa. - When choosing to pay upon receipt, go to the order summary page.

	- When choosing a visa, a page appears for him to fill in his visa information.
Outputs	The payment card will be added to the cards screen.
Constraints	- Successful login. - Valid card info.
Exceptions	- Wrong login info. - Invalid card info.

4.4.2 Provider Use Cases:

4.4.2.a Create a New Account Use Case (Signup)

When creating an account for the first time for the provider on the dashboard, he fills the requirement data, such as (name, email, address, password and verification code) and then he can use the dashboard.

Table 4.8 Sign Up Use Case

ID	7
Name	Create a New Account.
Description	The provider can register for the first time through the dashboard to be able to use it and add the products he wants.
Priority	High.
Actors	Provider.
Pre-conditions	An invitation code has been sent to the provider on his email.
Inputs	Name, email, address, password and invitation code.

Normal flow of events	- The provider enters the name, email, address, password and invitation code. - The home screen and the provider starts adding products. - Provider profile is displayed and can be modified.
Otherwise, Flow	- The dashboard cannot connect to the server because there is no internet connection or the provider entered the wrong login information. -The dashboard displays an error message.
Post-Conditions	-The dashboard displays the main screen. - Displays the categories to be filled, the description, and the images that he wants to display. - Viewing the profile of the provider, where he can modify or delete some of his information.
Outputs	None.
Constraints	- Successful login. - Valid product info.
Exceptions	- Wrong login info. - Invalid product info.

4.4.2.b Login Use Case.

When logging in to the provider in the dashboard, he fills in the data such as (e-mail and password) and then he can use the dashboard.

Table 4.9 Login Use Case

ID	8
Name	Login.

Description	The provider can enter through the dashboard to be able to use it and add the products he wants by filling in the data correctly.
Priority	High
Actors	Provider
Pre-conditions	Registered correctly
Inputs	Email, address, password.
Normal flow of events	- The provider enters the name, email, address, password and invitation code. -Pop-ups to the home screen and the provider starts adding products. - Provider profile is displayed and can be modified.
Otherwise, Flow	- The dashboard cannot connect to the server because there is no internet connection or the provider entered the wrong login information. -The dashboard displays an error message.
Post-Conditions	-The dashboard displays the main screen. - Displays the categories to be filled, the description, and the images that he wants to display. - You are viewing the profile of the provider, where he can modify or delete some of his information...
Outputs	None.
Constraints	- Successful login. - Valid product info.
Exceptions	- Wrong login info. - Invalid product info.

4.4.2.c Add Product Use Case.

The provider can add a product through the dashboard, where he also adds a description and several images of the product.

Table 4.10 Add Product Use Case

ID	9
Name	Add product
Description	The provider can add a new product through the dashboard to appear to the user.
Priority	High
Actors	Provider
Pre-Conditions	-Provider must be logged in first.
Inputs	Sub category, name (English, Arabic), info (English, Arabic), price, quantity and product images.
Normal Flow of Events	-The Provider enters sub category, name (English, Arabic), info (English, Arabic), price, quantity and product images. - New product is added in both sides (customer, business owner) list.
Otherwise Flow	- The provider has entered wrong login info. - The provider has entered wrong product info.
Post-Conditions	- New product is added in both sides (customer, business owner) list. - The provider can edit the product. - The provider can delete the product.
Outputs	- New product is added in both sides (customer, business owner) list.

Constraints	- Successful login. - Valid product info.
Exceptions	- Wrong login info. - Invalid product info.

4.4.2.d Update & Delete Product Use Case.

The provider can also update or delete the product through the dashboard, where the images, price, description or language can be modified, so the update or deletion is done from both sides on behalf of the customer and the provider.

Table 4.11 Update & Delete Product

ID	10
Name	Update and delete.
Description	The provider can update and delete product through the control panel.
Priority	Medium
Actors	Provider
Pre-Conditions	-Provider must be logged in first.
Inputs	Sub category, name (English, Arabic), info (English, Arabic), price, quantity and product images.
Normal Flow of Events	-The Provider can update any product info (sub category, name (English, Arabic), info (English, Arabic), price, quantity and product images) or delete the whole product. -The product is updated or deleted in both sides (customer, business owner) list.

Otherwise, Flow	- The provider has entered wrong login info. - The provider has entered wrong product info.
Post-Conditions	-The product is updated or deleted in both sides (customer, business owner) list.
Outputs	-The product is updated or deleted in both sides (customer, business owner) list.
Constraints	- Successful login. - Valid product info.
Exceptions	- Wrong login info. - Invalid product info.

4.4.2.e Create Category Use Case.

The provider can add more than one category through the dashboard, where he names the item and puts all its details. If he enters incorrect information, he cannot display the new category.

Table 4.12 Create Category Use Case

ID	11
Name	Create category.
Description	The provider can add more than one category through the control panel.
Priority	High.
Actors	Provider.
Pre-Conditions	-Provider must be logged in first.
Inputs	Category name (english, arabic) and category image.

Normal Flow of Events	-The Provider can enter the category name (english, arabic) and category image. -The category is added to both sides (customer, business owner) list.
Otherwise, Flow	- The provider has entered wrong login info. - The provider has entered wrong category info.
Post-Conditions	- The provider can add a sub-category to the main category. - The provider can add products within the sub-category. - The provider can edit and delete the main category. - The provider can delete and edit the sub-category.
Outputs	-The category is added to both sides (customer, business owner) list.
Constraints	- Successful login. - Valid category info.
Exceptions	- Wrong login info. - Invalid category info.

4.5 Sequence Diagrams

In this section we display a sequence diagram which is an interaction diagram that details how operations are carried out.

Figures 4.4, 4.5, and 4.6 show the data flow in the login process, order, and add product respectively.

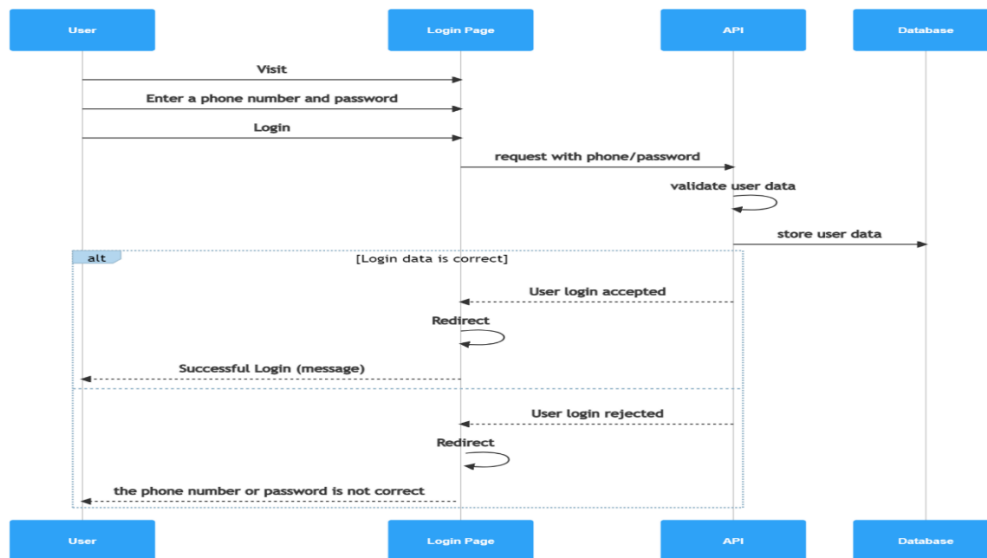


Figure 4.4 Login Process

Figure 4.5 show the data flow in the order process.

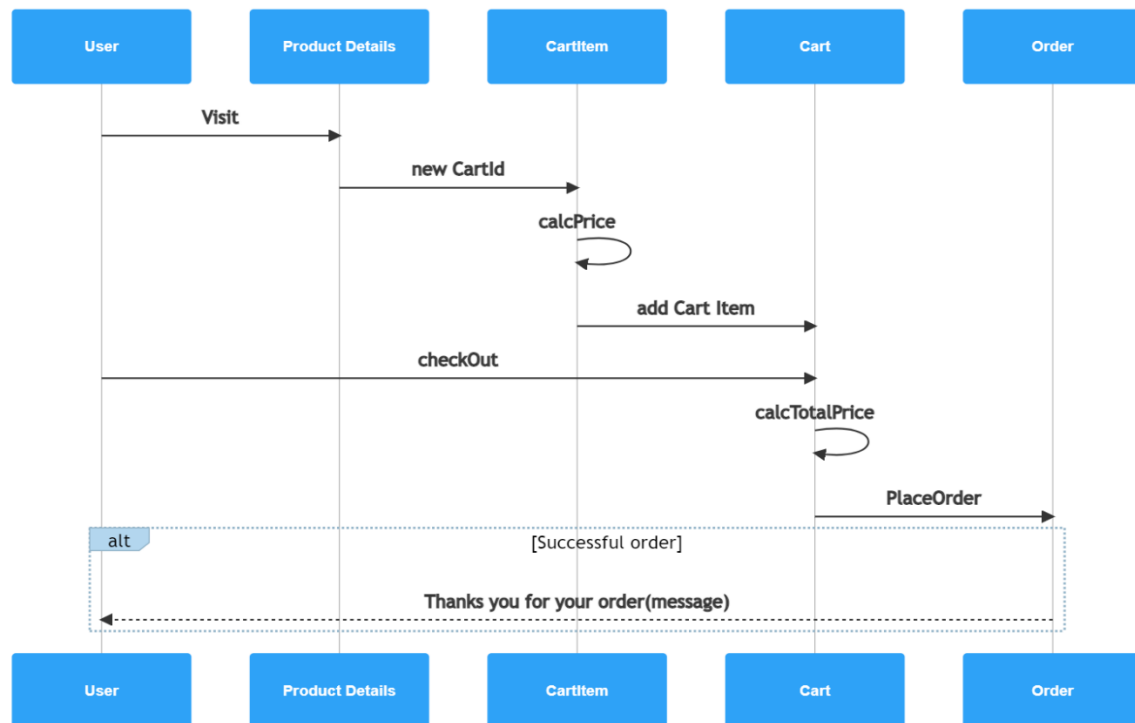


Figure 4.5 Order Process

Figure 4.6 show the data flow in the add product from provider.

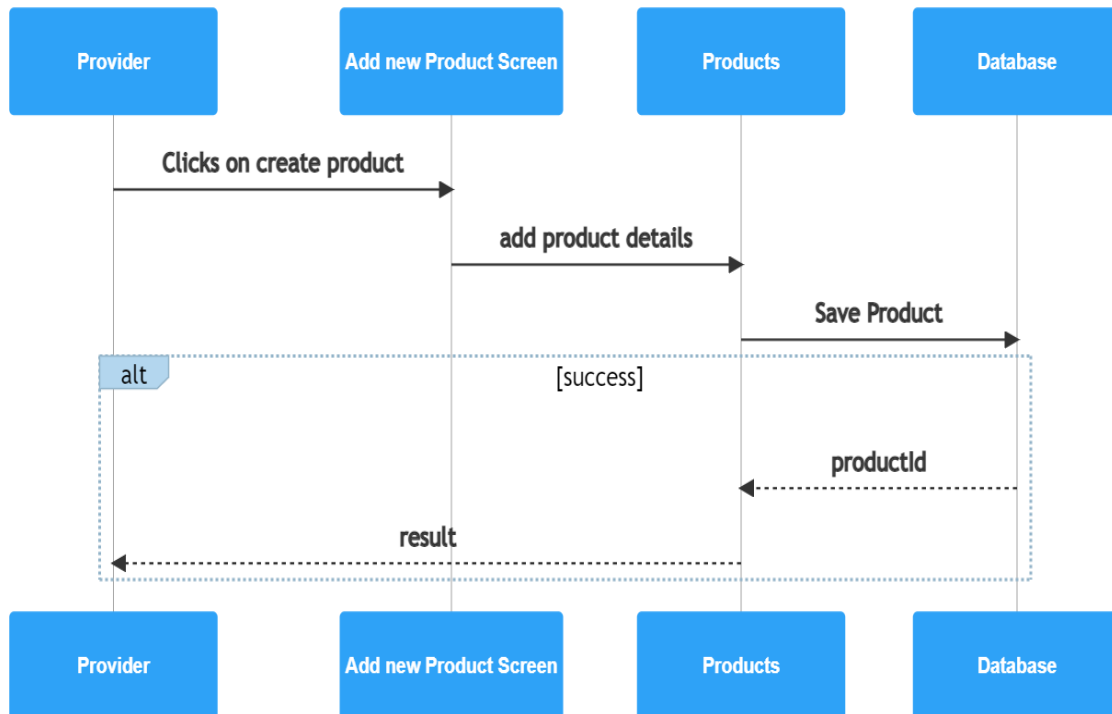


Figure 4.6 Add Product

4.6 Risk Management

Project Risk Analysis and Management is a crucial process that enables the analysis, assessment, and management of the risks associated with a project. Properly undertaken, it will increase the likelihood of successful completion of a project in time, within budget, and with high performance.

Table 4.12 Project risk management

ID	Risk	Type	Possibility of occurrence	Rank	Solutions
1	The number of visitors to the application is small according to the maximum required.	People	medium	High	Social media marketing.
2	Performance, quality.	Technology	Low	Moderate	Simulation, Modelling, Prototyping.
3	Hardware failures (Mobiles, laptops).	Technology	Low	Very High	Stand-by devices, Regular backup.
4	Technology Change.	Technology	Low	Low	Adapt the new technology as fast as possible.
5	The application is targeted to a specific country, which leads to the inability of users to access (this	Technology	Low	High	Provide the application for all countries and when you enter the phone number, a message appears. This application is not

	application is not available in your country).				available in your country.
6	Application performance in time taken to execute operations (load time) The process here is in performing the query.	Technology	medium	High	Through programming, the best query is used.
7	Withdrawal of a member of the work team (developers).	People	high	Moderate	The presence of alternative people in the work team to avoid this problem.
8	Exceeding some powers, for example, the user has no right to appear except for the data required for him.	Technology	low	low	Check their databases and queries before handing over the project thoroughly.
9	The inability of all users to apply to some devices.	Technology	Low	Low	Providing all versions on all devices (Android, iPhone, desktop).

Chapter 5

Design

5.1 Introduction

In this chapter, We start by to explain the structure of the application and then we drew the entity relationship diagram that describes the data database table, then we explained the user experience of the application and the description of the application, then we defined the standards of the user interface, and at the end of that, we put the initial drawing of the application and the design of the control panel, as well as the final design of the interfaces.

5.2 Information Architecture (IA)

Information architecture in terms of planning and UX design can be considered as the science of structuring and organizing the content in websites and mobile applications. It is basically a practice of deciding how to arrange the application parts to make it understandable and easy to use and find anything the user may need [12].

Figure 5.1 shows the information architecture map for Pal-Pazzar App.

Pal-Pazzar App

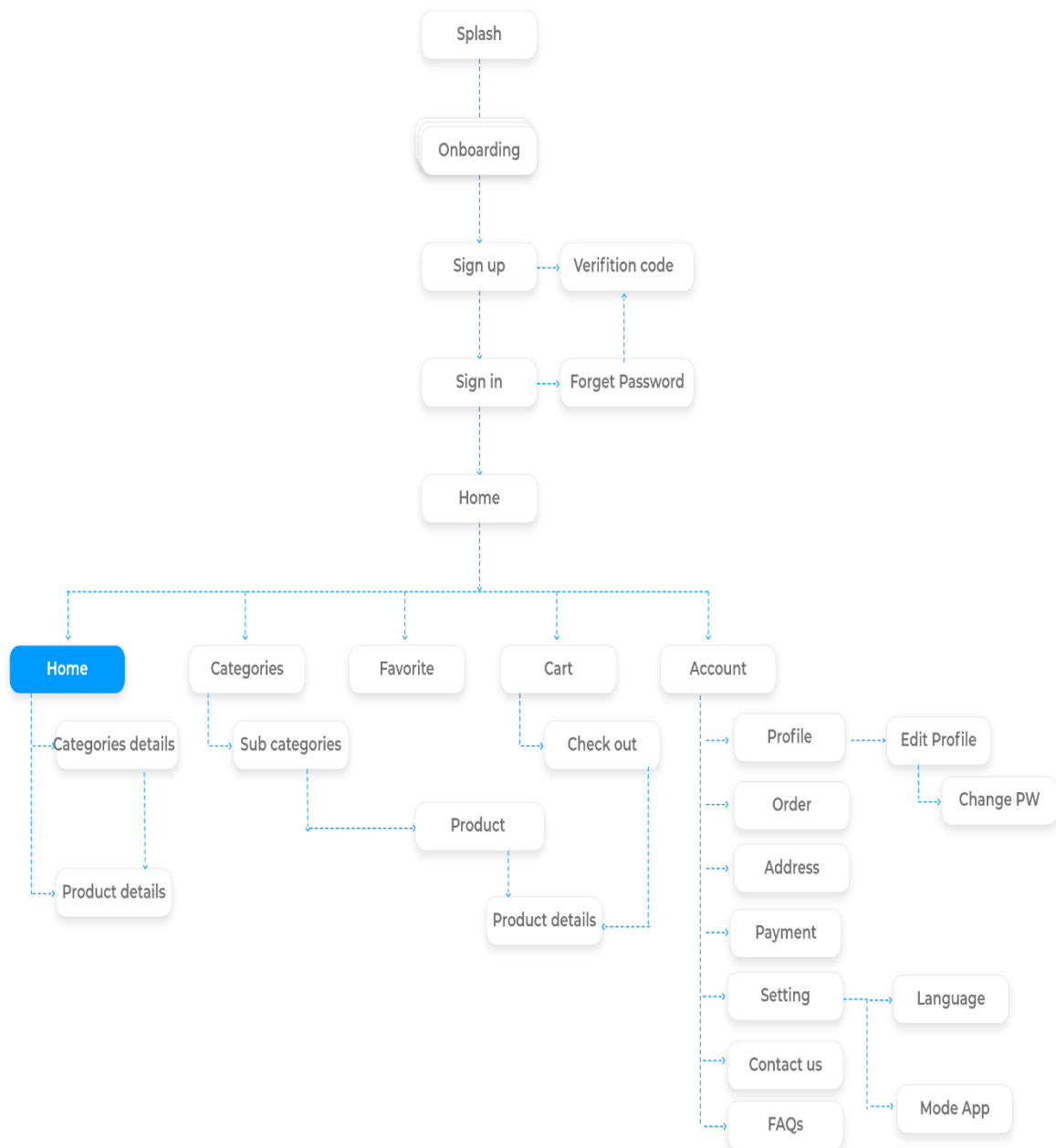


Figure 5.1 Information Architecture Map

5.3 ER Diagram

Entity Relationship (ER) Diagram Describes database tables, fields, and relationships between entity sets.

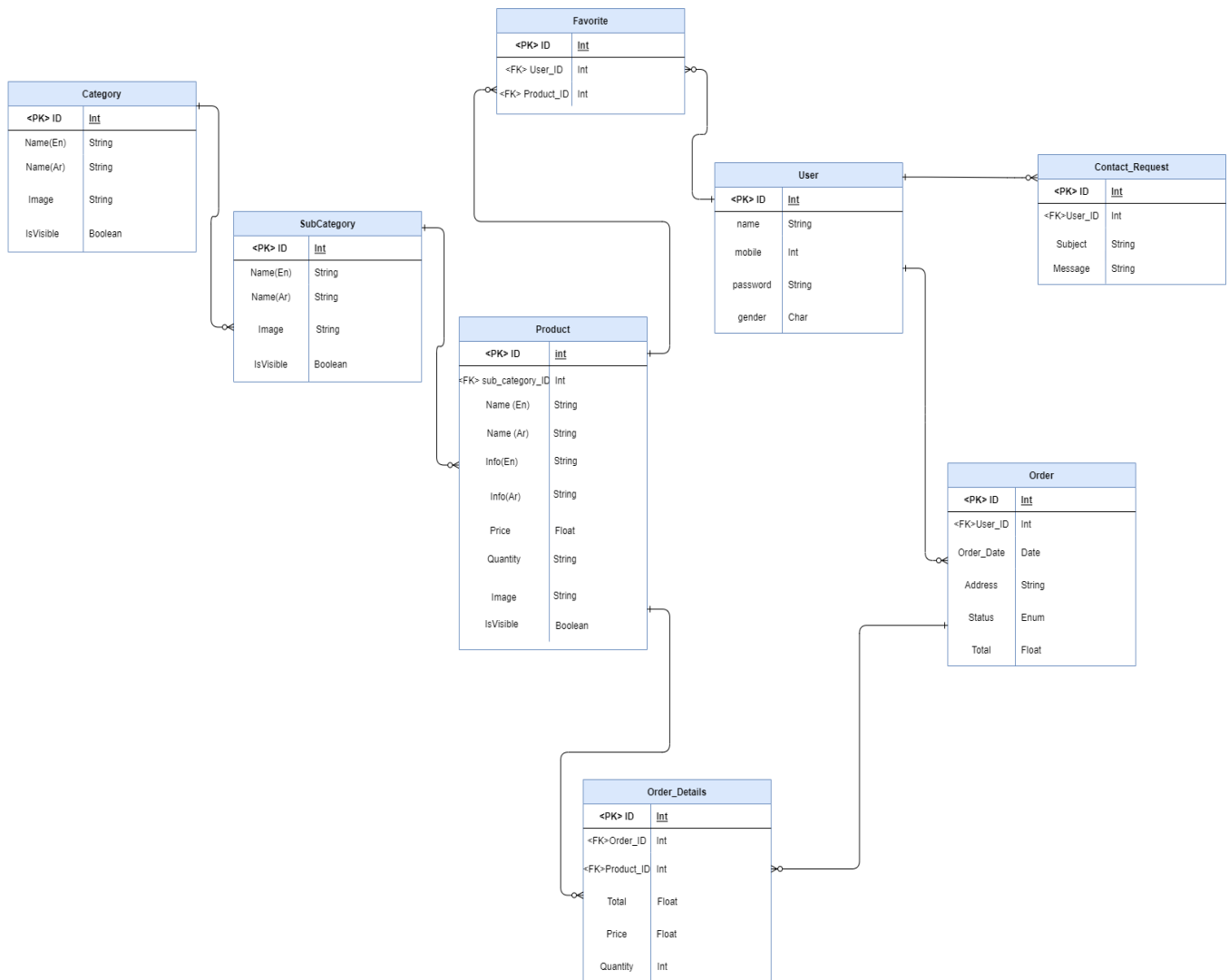


Figure 5.2 ER Diagram for our system.

5.4 UX & UI Analyses

App Name:

Pal-Pazzar.

Idea:

The application is a store to display the pioneering projects of Palestinian products and sell them in several categories.

Description:

The application allows the user to purchase the product that he likes at also sees newly added and best-selling products. The application show the status of the order and provider different payments method.

Example:

I am a store owner and I want an application that allows me to easily display my products to the users and browse them all, and also allows me to add products and display them clearly and update all new, the store owner presents his own products to the main application owner and the owner adds the products of the store owner in the Dashboard within the agreed upon.

4.5.1 The difference between the icon and the button

- *An icon is representative of something. You can have a company newsletter with icons in the margin of each story representing IT, Sales, Marketing, etc.
- *A button is by definition interactive. You click on it and something happens. It may have pictures, words, or both [13].

4.5.2 UI required:

1. Splash

This screen contains the app icon/logo and appear for 3second only.

2. Onboarding

[Several explanatory screens for the user to move directly to the application]

- Action: (skip button and next button)

3. Registration

[The account creation screen enables the user after registration to use all the features of the application]

- Input :(Name , number, password, gender, and city).
- Action :(Register button).

4. Verification code

- Input: (Enter numbers of verification code)
- Action: (Verify button)

5. Forget Password

[Screen appears to the user in case the password is forgotten]

- Input: (Phone number)
- Action :(Send button)

6. Sign In

[Screen appears to the user if he has an account before]

- Input: (Phone number, password)
- Action: (Forgot password, login button)

7. Home

[The main screen in the application and contains several contents]

- Action :(Search, App mode, add to favorites)

8. Categories

[Screen showing the categories inside the application]

9. Product Details

[The product details screen shows all the information of the product]

- Action :(Add to favorites, add to cart)

10.Sub categories

[Screen that contains the sub-categories of categories]

- Action: (Add to favorites, Transfers between items)

11.Favorite

[Screen that contains the products that added by the user]

- Action: (Remove the product from your favorites , product details)

12.Cart

[Screen contains the list of product that added by the user in order to purchase it]

- Input: (Quantity)
- Action: (check out)

13.Check out

[Screen which the user can purchase the product in the way that suits him]

- Input: (All the information you need for your credit cards or your information)

- Action: (Choose the payment method, check out button)

13.1 Cash on delivery

[When choosing this method, go to the order details page]

- Action: (Check out button)

13.2 Visa Card payment method

[When choosing this method, he goes to the appropriate visa selection page for him such as: Card ID, CVV, Expiration date]

- Action: (Check out button)

14.Profile

[Screen containing many icons that must be included in the application]

- Action: (update profile, settings, contact us, order, address, payment, FAQs)

15.Update Profile

[Screen that provide ability to the user to modify all his data]

- Input: (gender, name, number, city)
- Action: (Save change)

17. Setting

[Screen provide ability to the user to change language and mode App]

- Action: (language, change password , App mode)

18. Order

[Screen that contains customer information and includes the status of the order, the payment method and the price of the order]

- Action: (Check out button)

19.Contact us

[Screen provide ability to communicate with the store owner]

- Input: (subject, message)
- Action: (send button)

20. Change Password

[Screen appears when you want to change your password]

- Input: (Old password, new password, confirm new password)
- Action: (Save button)

21. Address

[The screen provides the ability to show all the details of the user's address and he can edit, delete or add address]

- Input: (name, city, street, phone number)
- Action: (Edit button, Delete button, Add button)

22. FAQ

[Screen showing frequently asked questions from users]

- Action: (Choose a question and see more details about the App)

23. Reset Password

[A screen that provides the user with the ability to reset his password when he forgets it]

- Input: (Verification code , new password, confirm new password)
- Action: (verify button)

4.5.3 App Color

It is said: “Colors have strength and energy” and it is undoubtedly true, as colors really affect our moods, emotions, and behaviors as well. It can also be a source of information sometimes. It is generally not easy to build a simple and adequate design, your focus as an interface designer will be on the usability, consistency and visual distribution of the elements [14].

We chose the orange color for the application because it gives a feeling of warmth, also gives a sense of optimism, enthusiasm and passion and expresses activity and energy.

5.5 Design

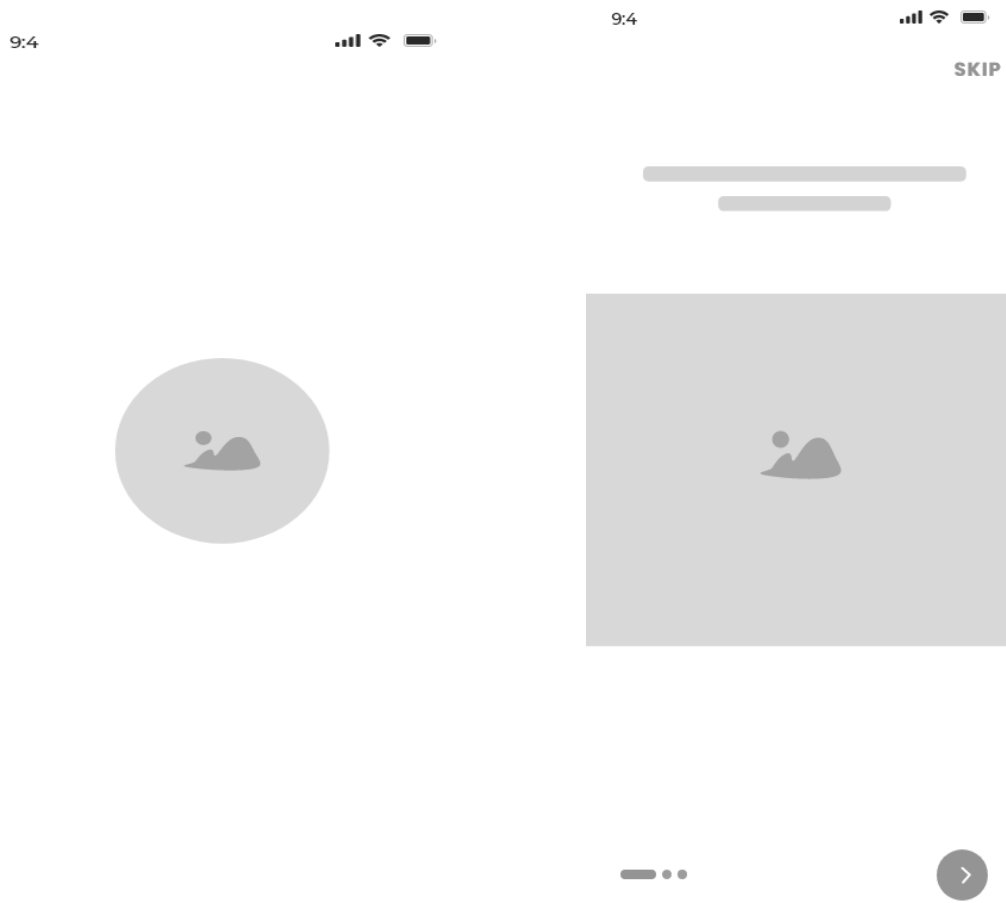
5.5.1 Adobe XD

Is a vector-based tool developed and published by Adobe for the design and experience of web user design, for web and mobile applications. The program is available for Mac OS and Microsoft Windows. It is one of the distinguished programs for designing various types of user interfaces, streamlined and efficient interface Responsive design testing Linking elements together and product testing Fixed, absolute determination of transition effects value[15].

5.5.2 Wireframe

A wireframe is a simple design draft as an initial form that shows the client how to structure the site. It contains onboarding screens, registration screens, forget password screen, and verify password, and contains my account page and some important screens that need to be understood, such as the shopping cart or categories inside the application and the most important page is the home page and the payment

screen and its methods and the address screen and other screens are explained in details.



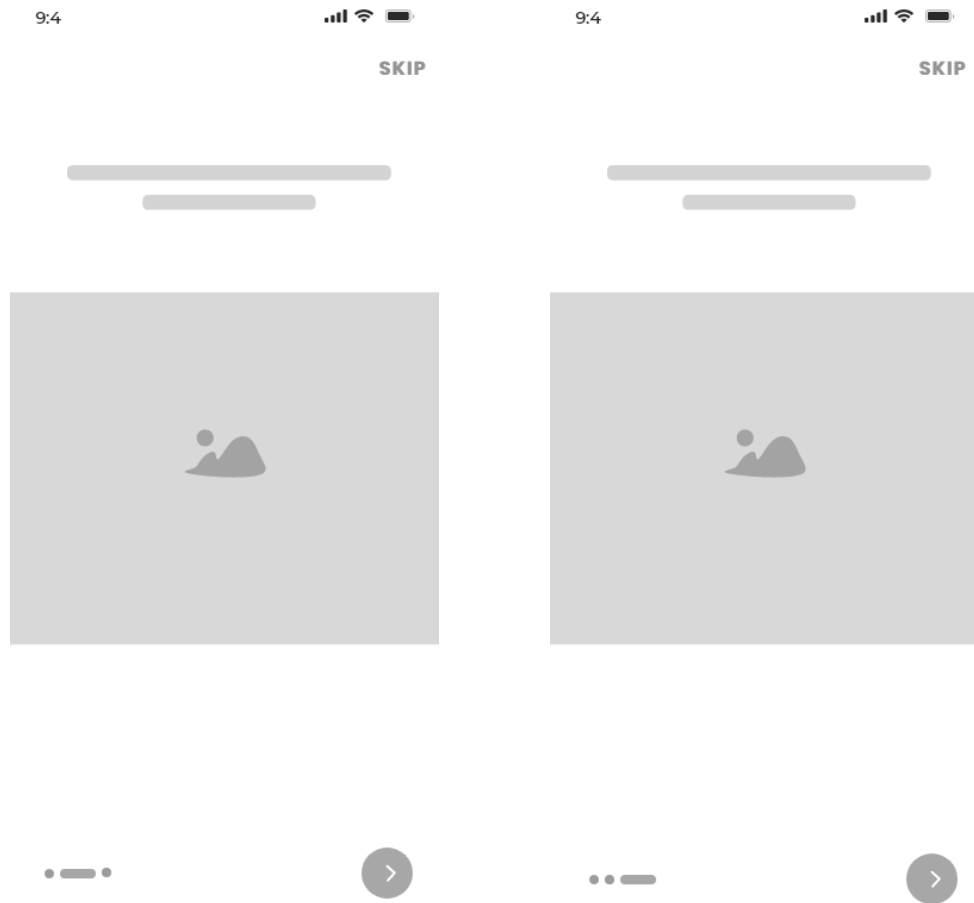


Figure 5.3 Launch Screens Wireframe.

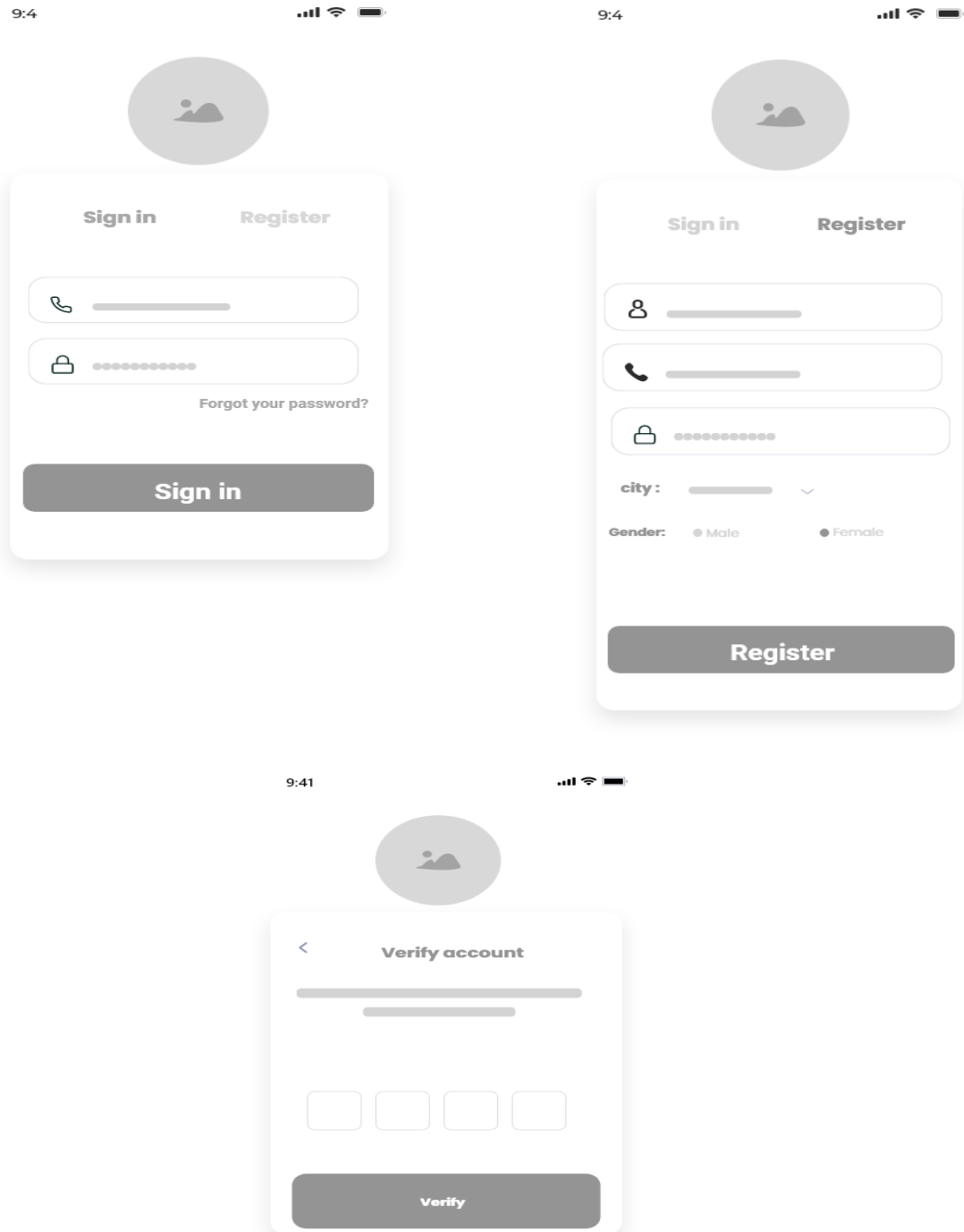


Figure 5.4 Register Screens Wireframe.

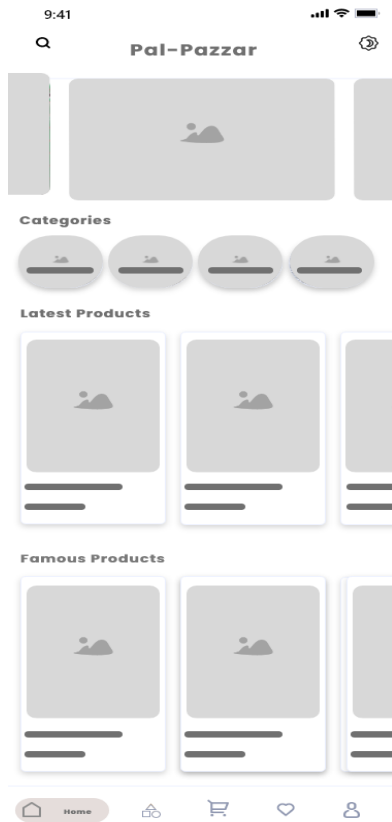


Figure 5.5 Main Screens Wireframe.

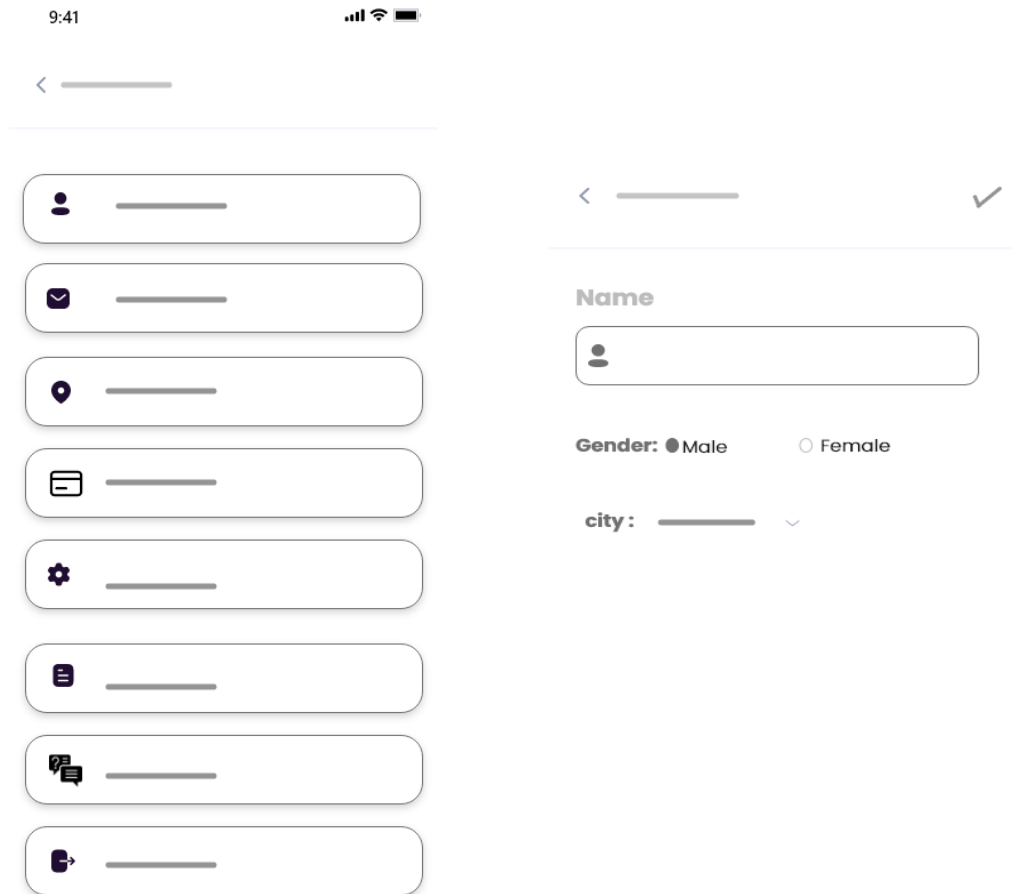


Figure 5.6 Account Screens Wireframe.

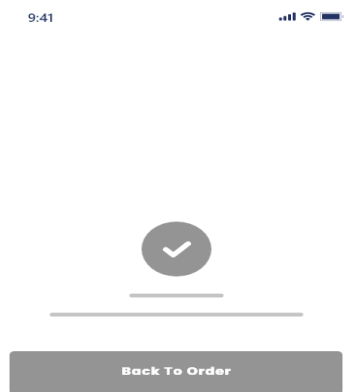
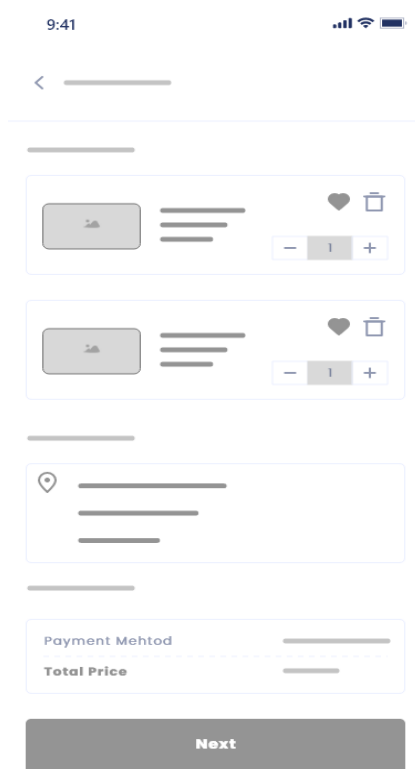
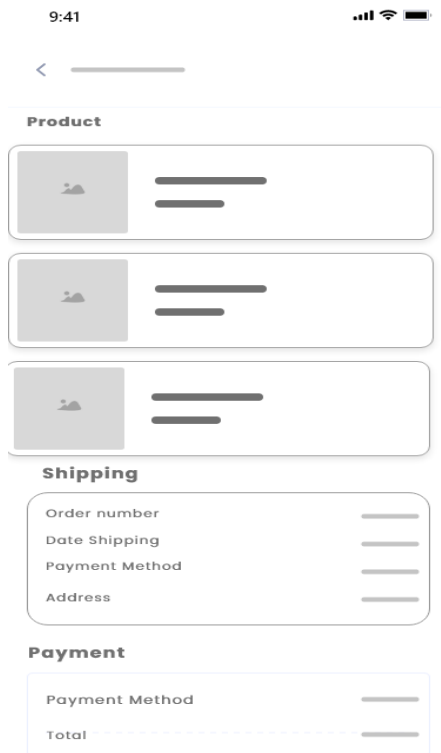


Figure 5.7 Order Screens Wireframe.

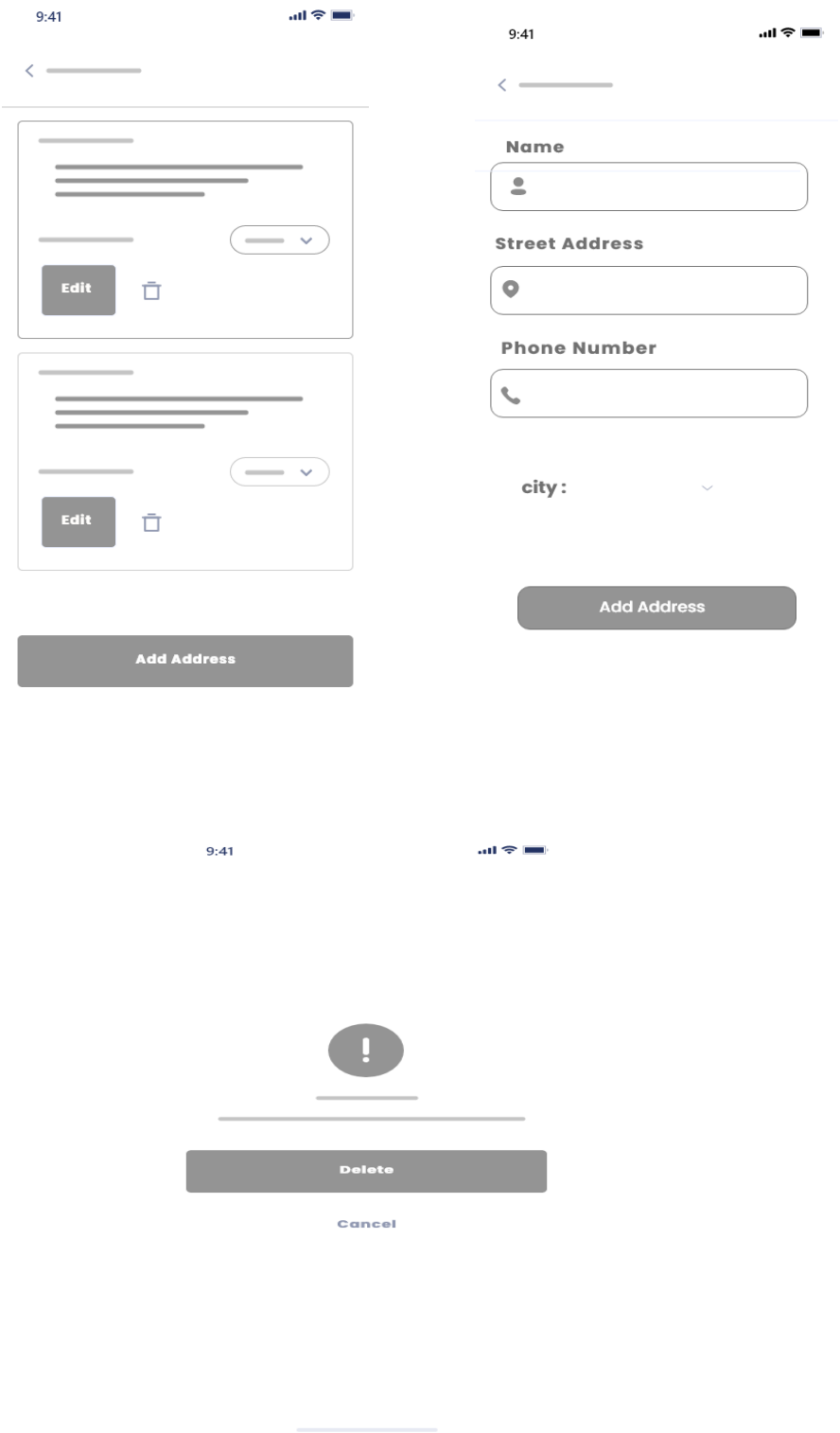


Figure 5.8 Address Screens Wireframe.

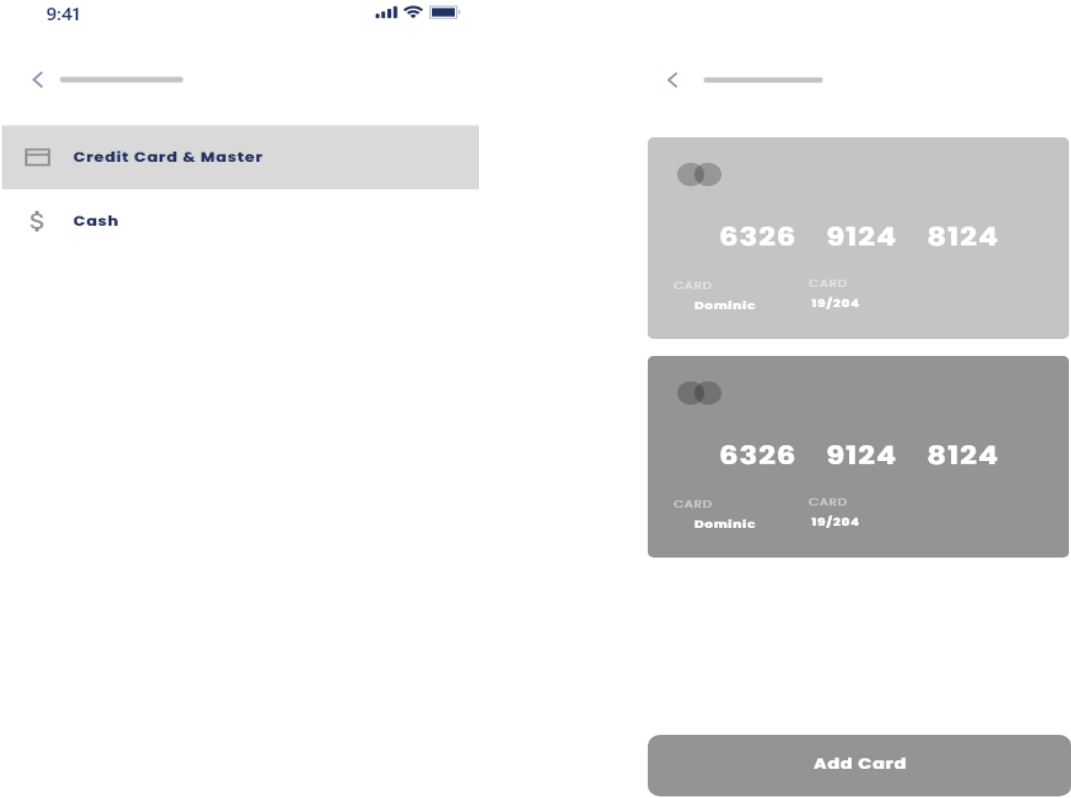


Figure 5.9 Payment Screens Wireframe.

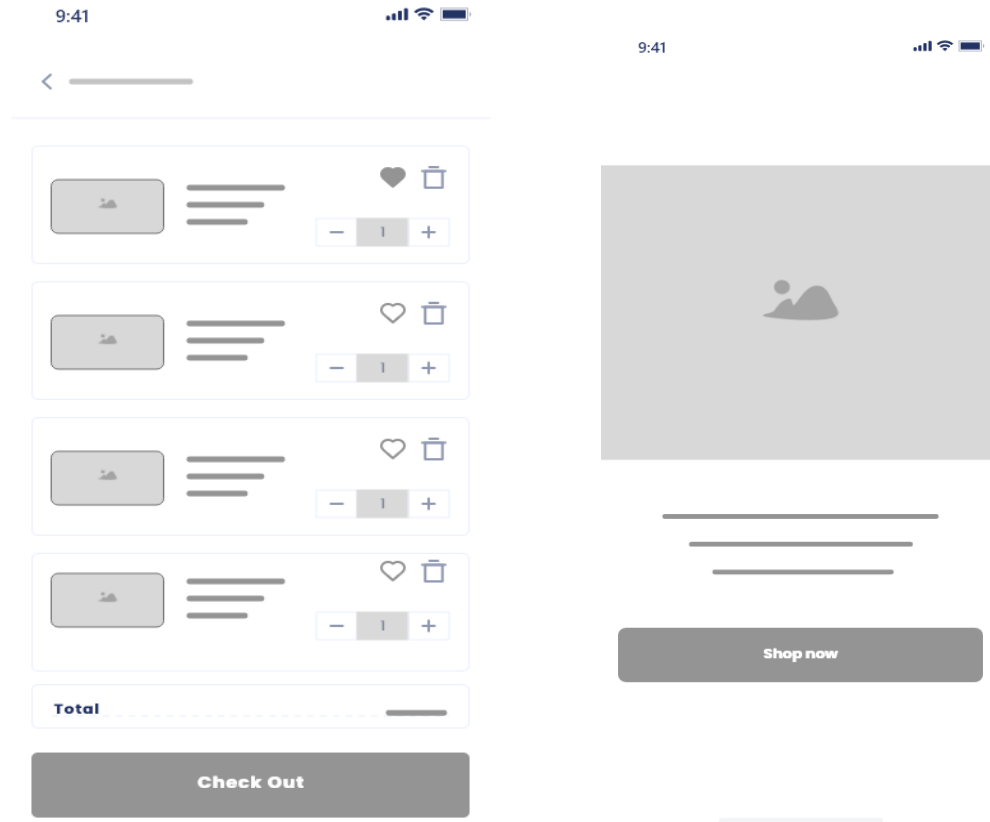


Figure 5.10 Cart Screens Wireframe.

5.5.3 Dashboard

Dashboard used to manage all process of business owner such as :(add products, categories, and subcategories) also the admin can receive any contact request from the users and can send notifications to specific user as well as see the total incomes, customers, and products[16].

Link:

<https://smart-store.mr-dev.tech/cms/store/login>

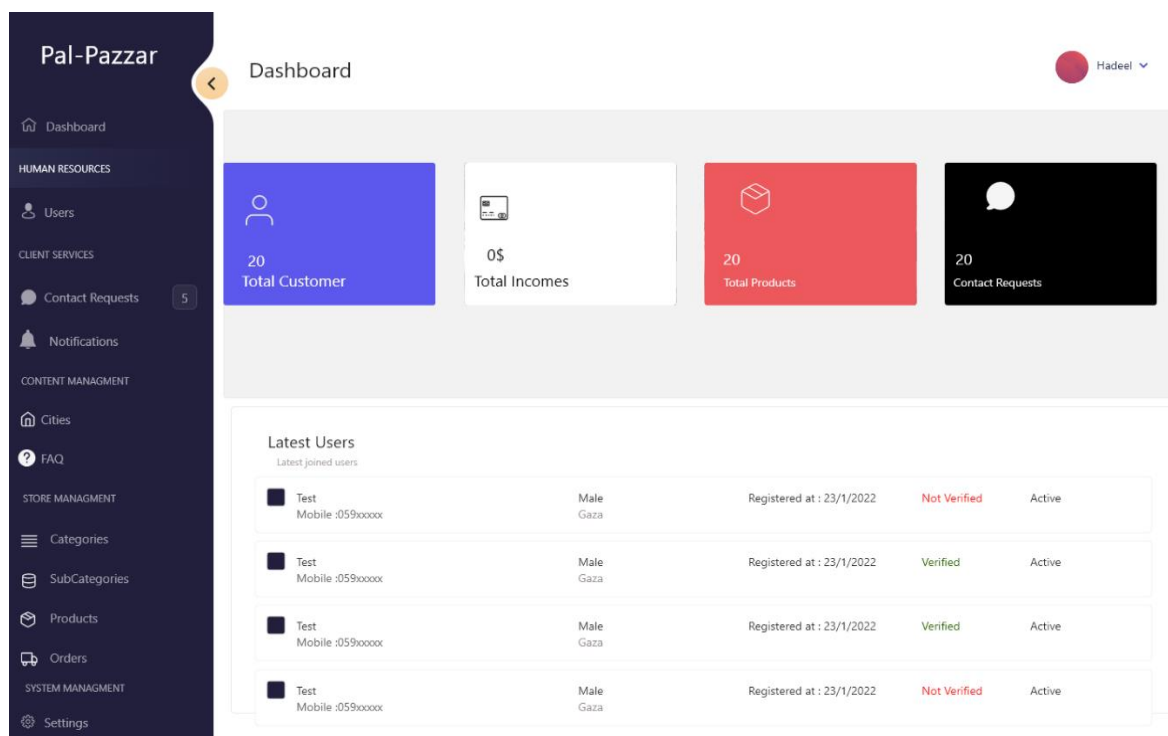


Figure 5.11 Dashboard of Business owner

5.5.4 UI Design

These screens start with the screen that contains the application logo and then the onboarding screens and contain the features offered by the application.

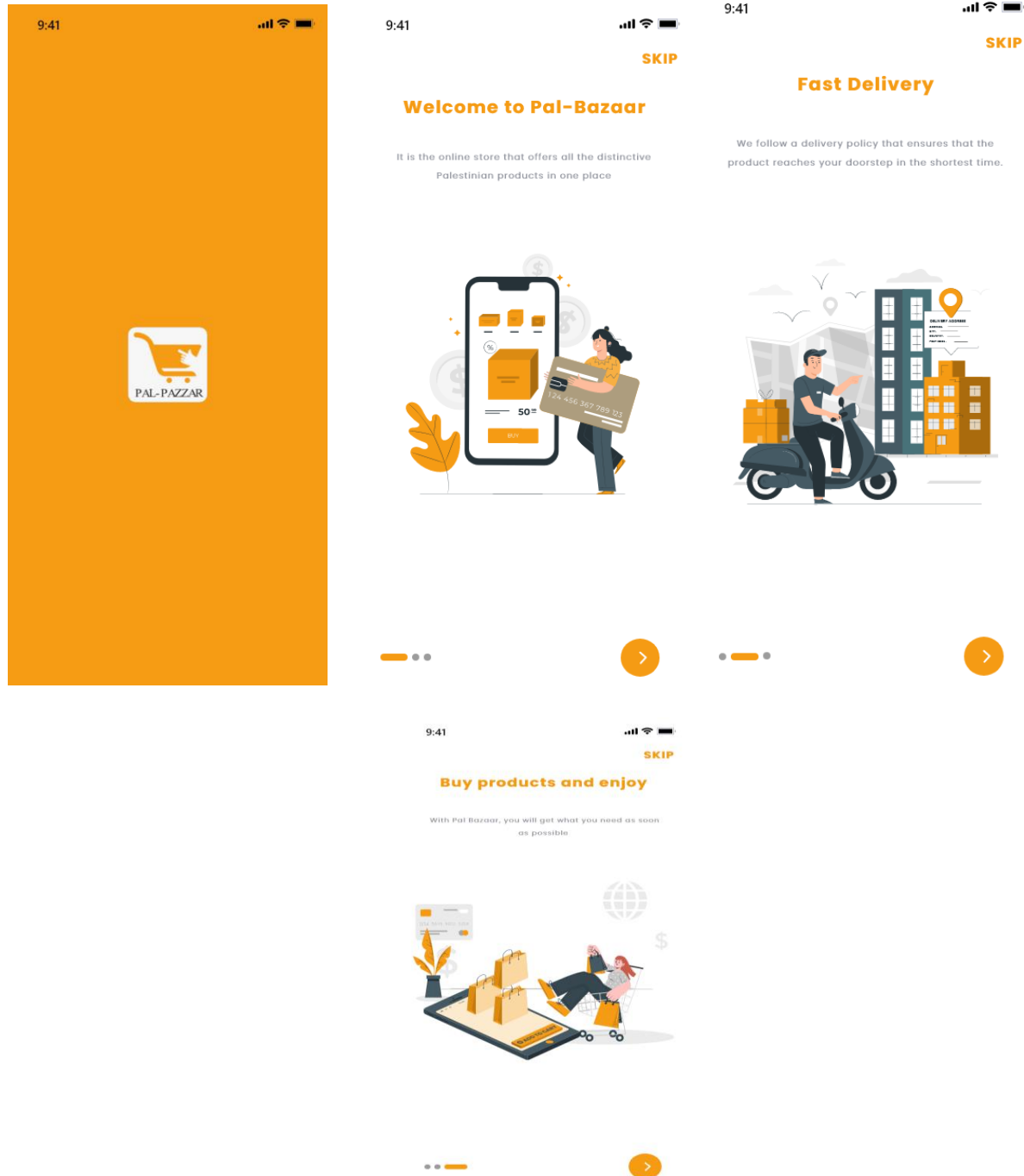
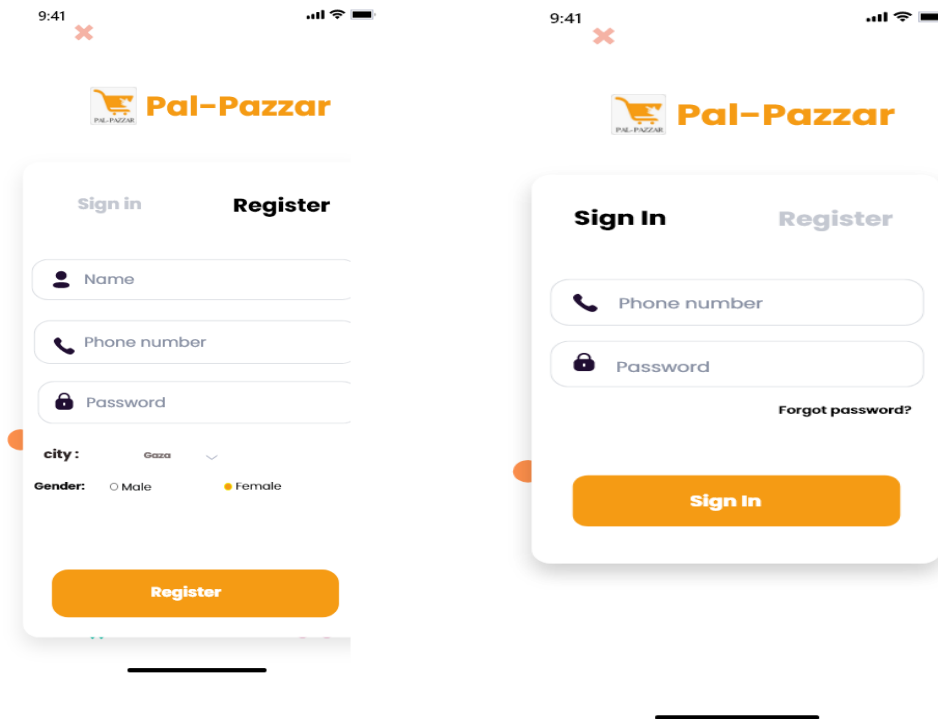


Figure 5.12 Launch Screen Design

And then the registration screens and it contains two screens, pre-registered or I do not have an account, so you fill in the required. The third screen appears about forgetting the password, the fourth screen to verify the password, it sends you a code that you enter in the box, the fifth screen to reset the password and sixth screen for change password.



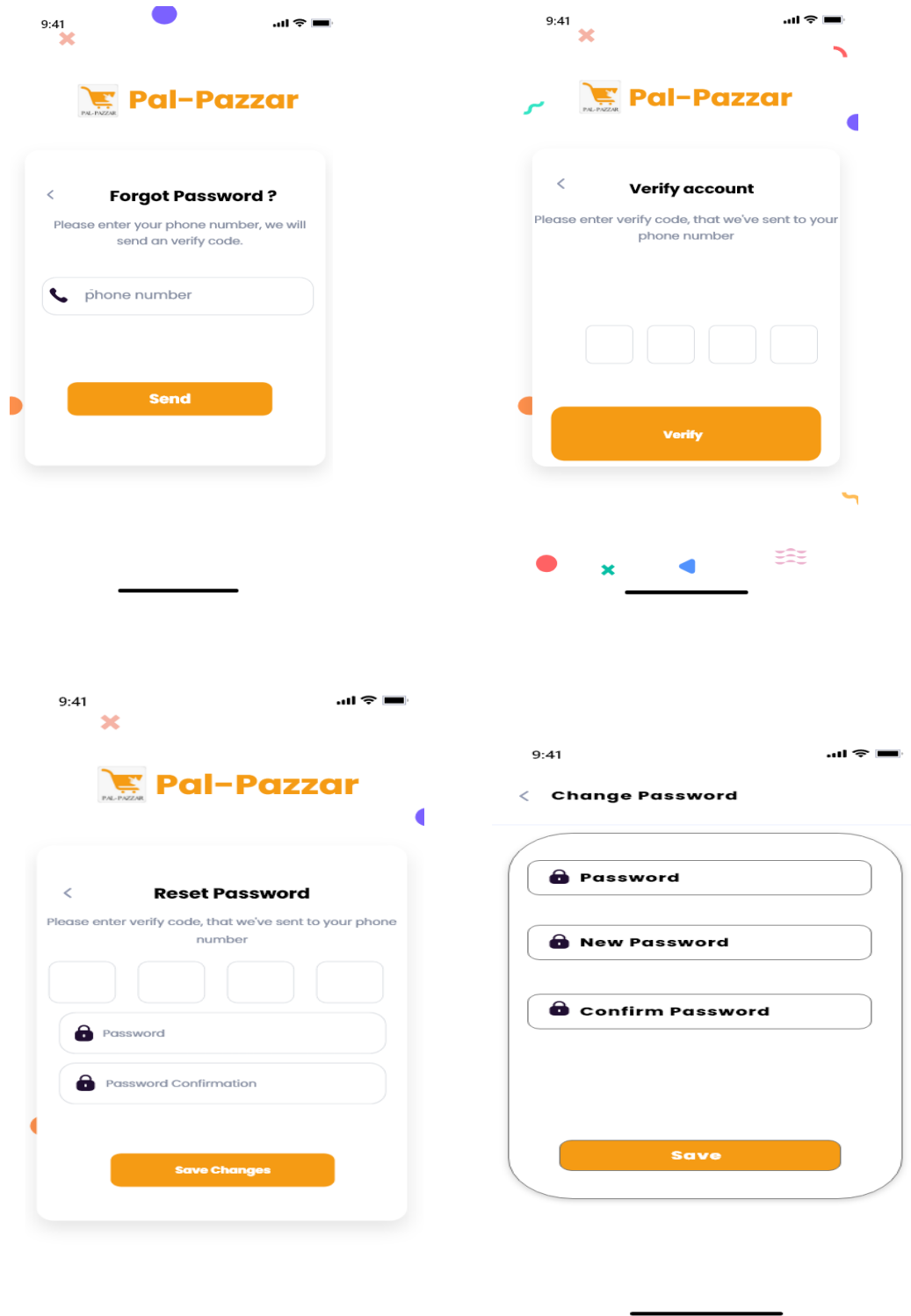


Figure 5.13 Register Screens Design

The first screen is the search screen for any product you want and it shows you the product that you searched for, and the second screen appears when you do a wrong search or a product that is not within the products.

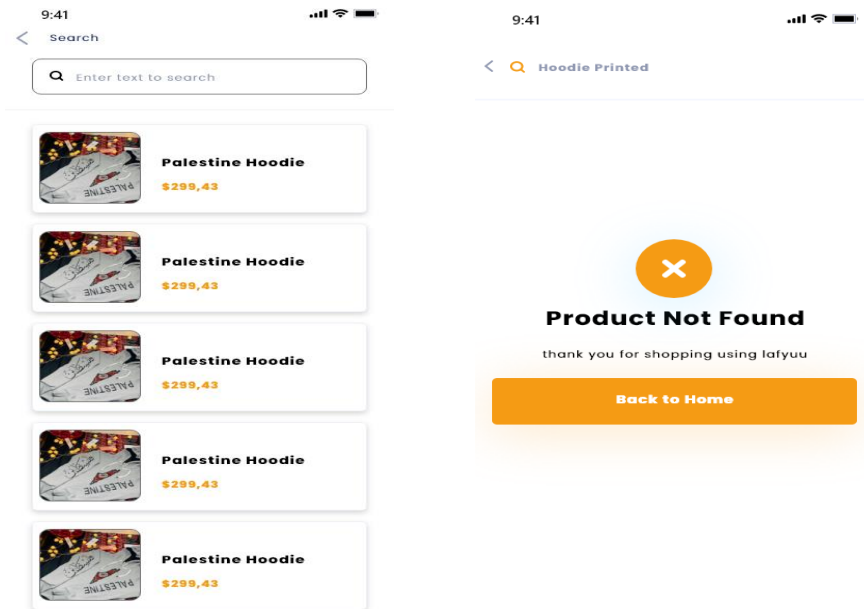


Figure 5.14 Search Screens Design

The settings screen of the App and it contains a change of password and setting the language of the App.

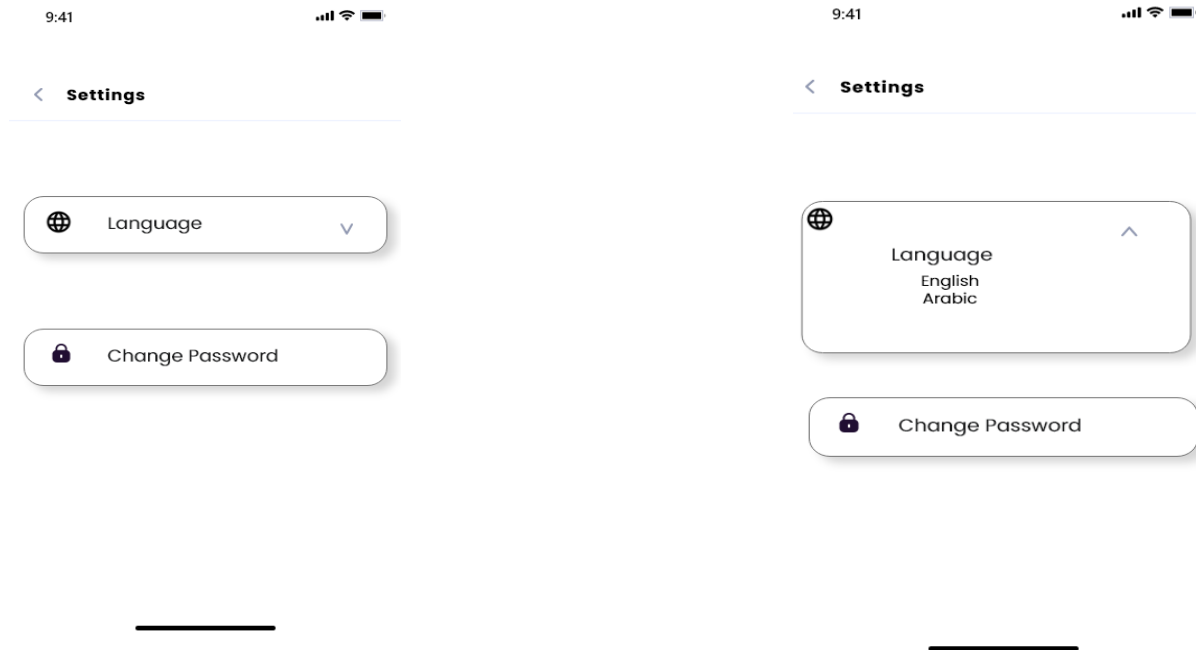


Figure 5.15 Settings Screens Design

The cart screen collects all the products you want to buy, while the second screen is that appears to you when the cart is empty.

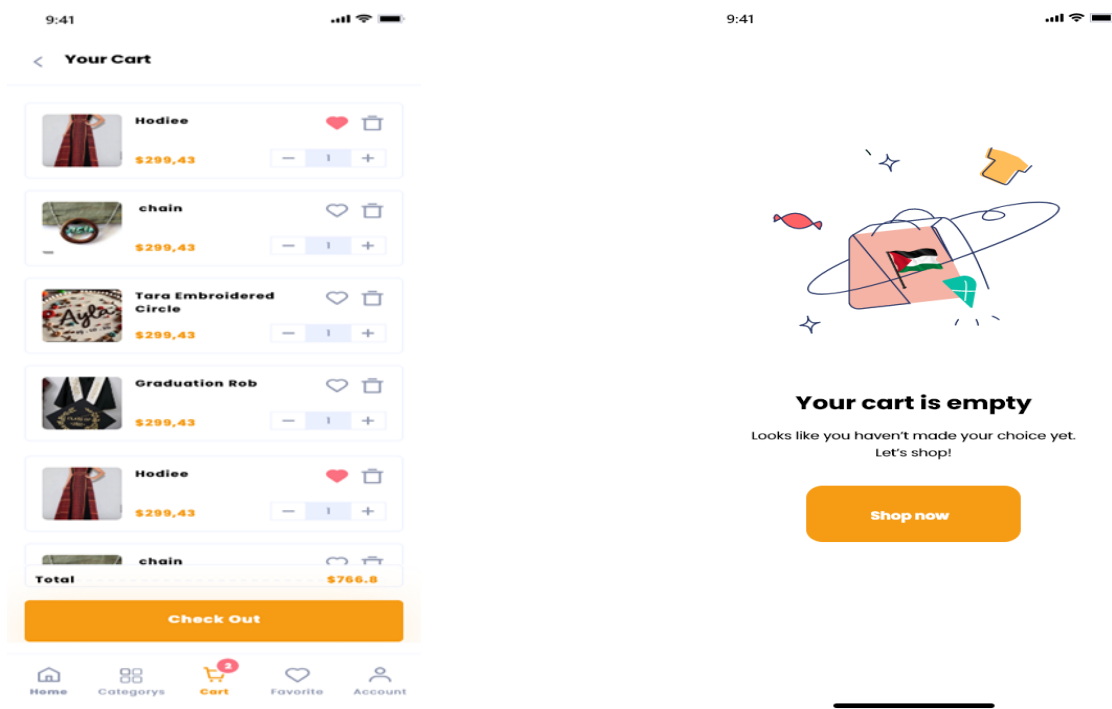


Figure 5.16 Cart Screens Design

Address screens appear to you when making a purchase, adding your address and filling in the required data, and you can modify, delete or add an address

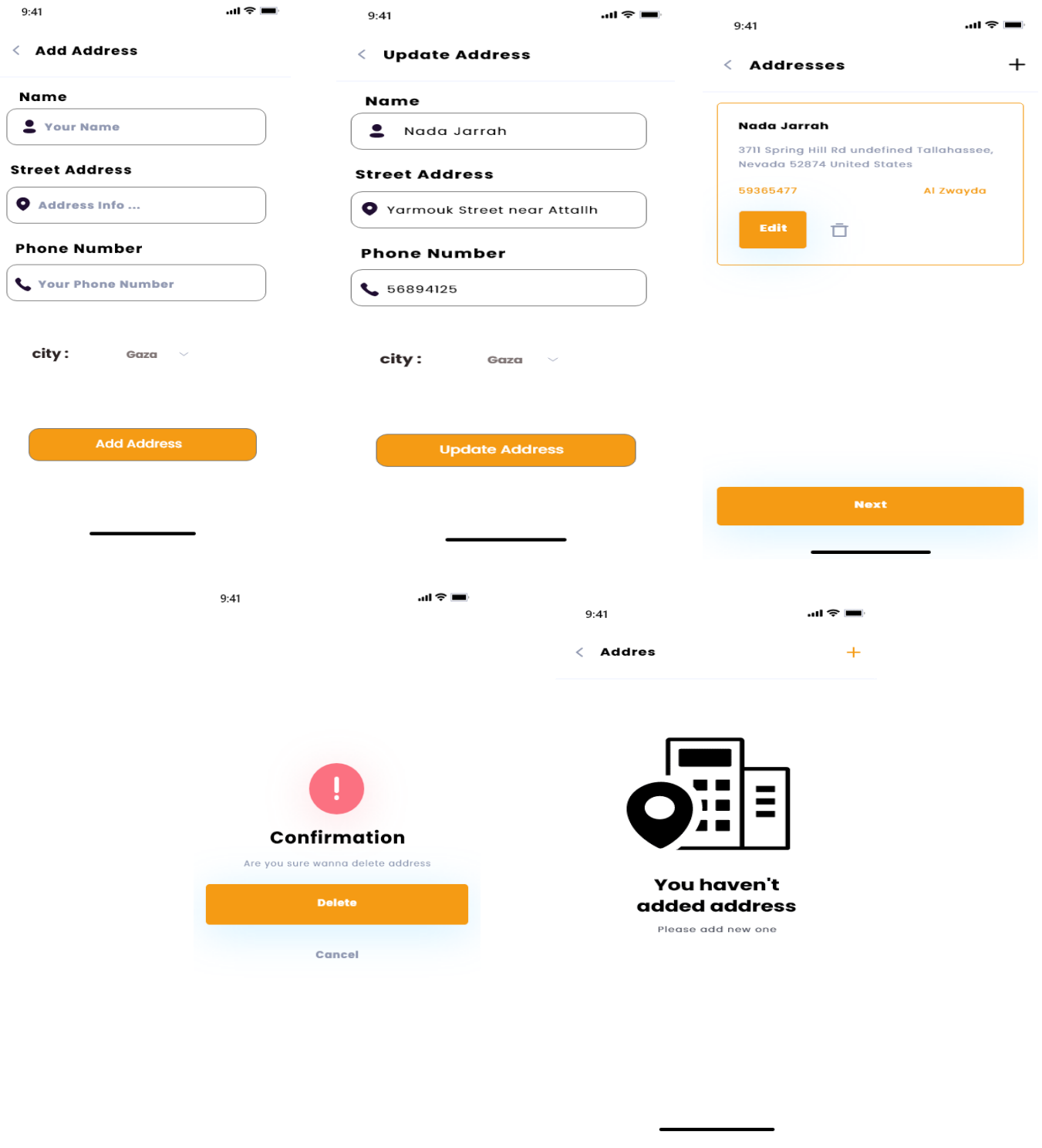


Figure 5.17 Address Screens Design

The first screen is the home page that contains several things like searches, categories, etc., while the second screen is the newly added products, and the third screen is the products you might like.

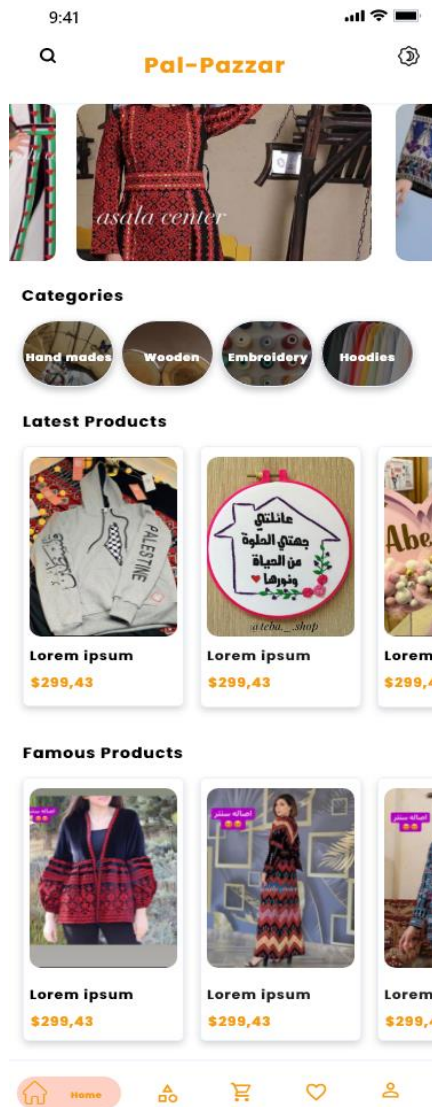


Figure 5.18 Main Screens Design

Screens for my account page, through which you can modify the profile page, and there is also a page for common questions and a page to contact us if you have problems.

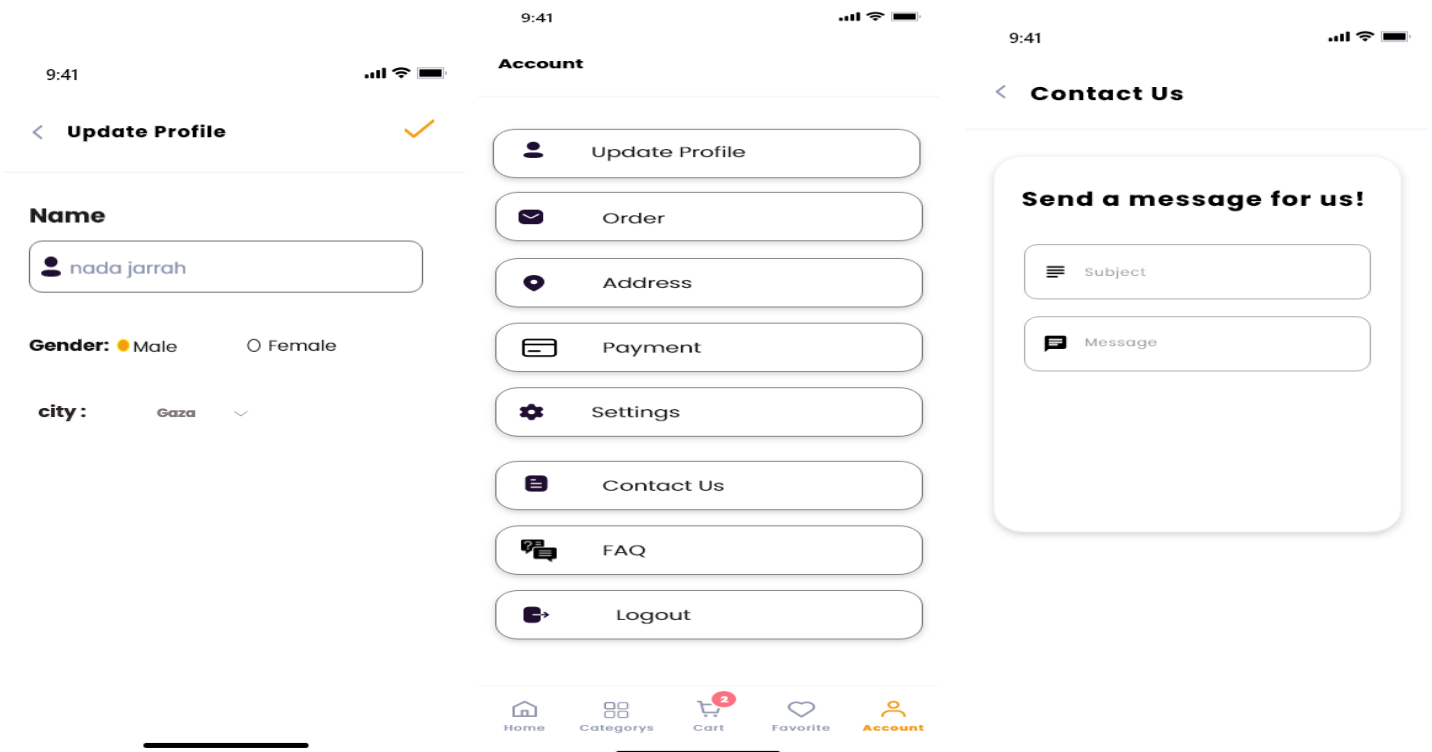
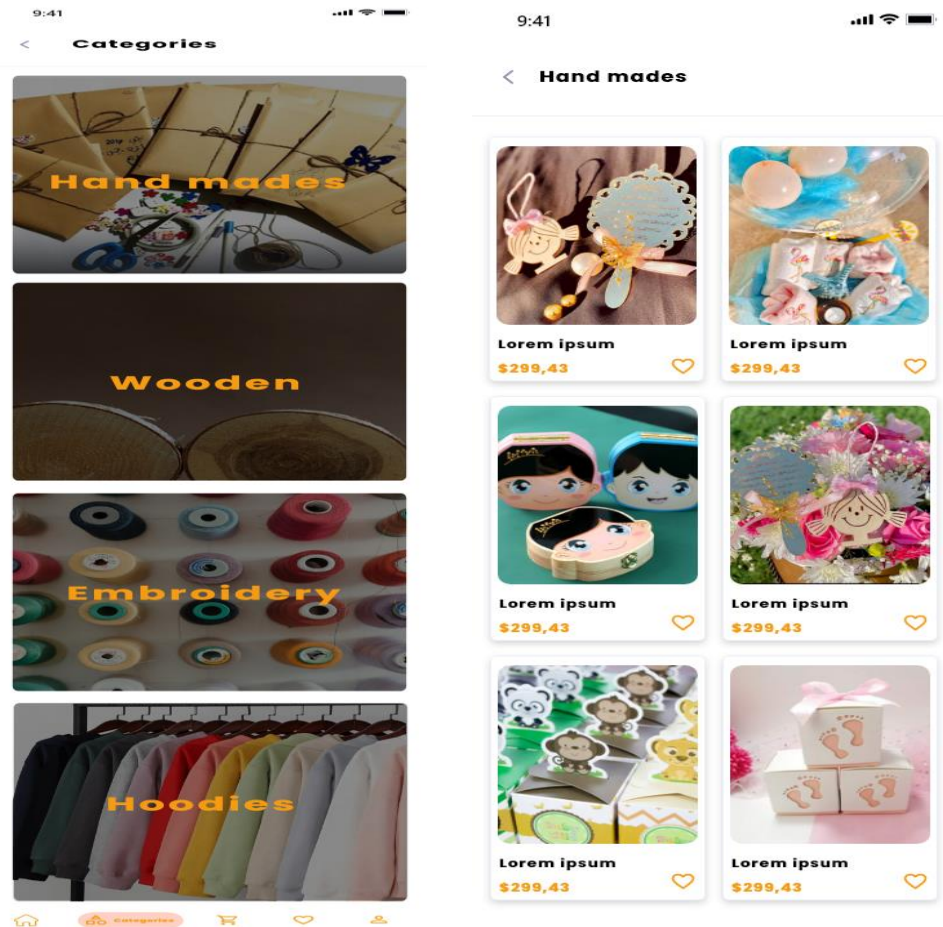


Figure 5.19 Account Screens Design

The main categories screen and each category contains a sub-category and the following screens show that.



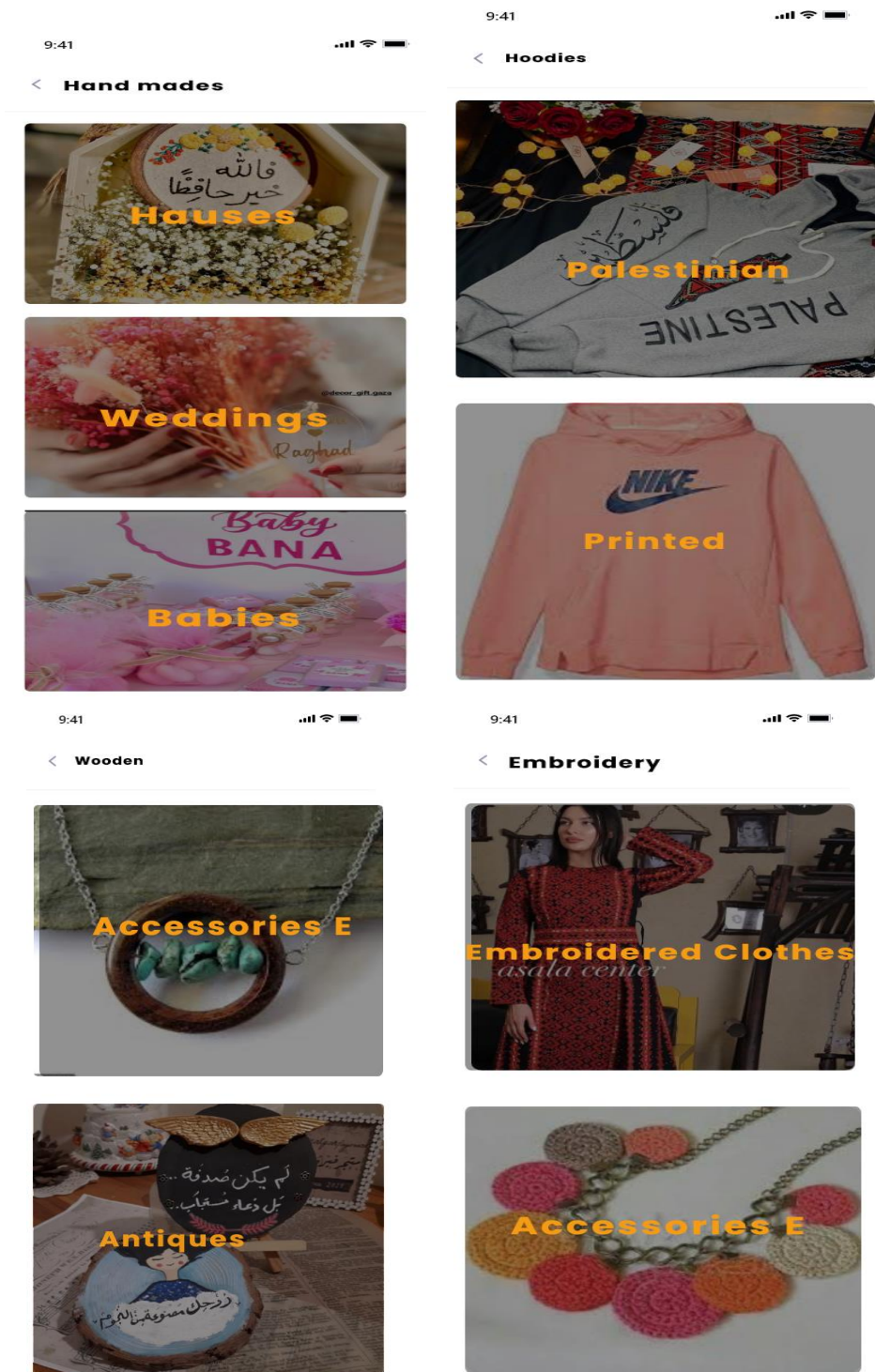
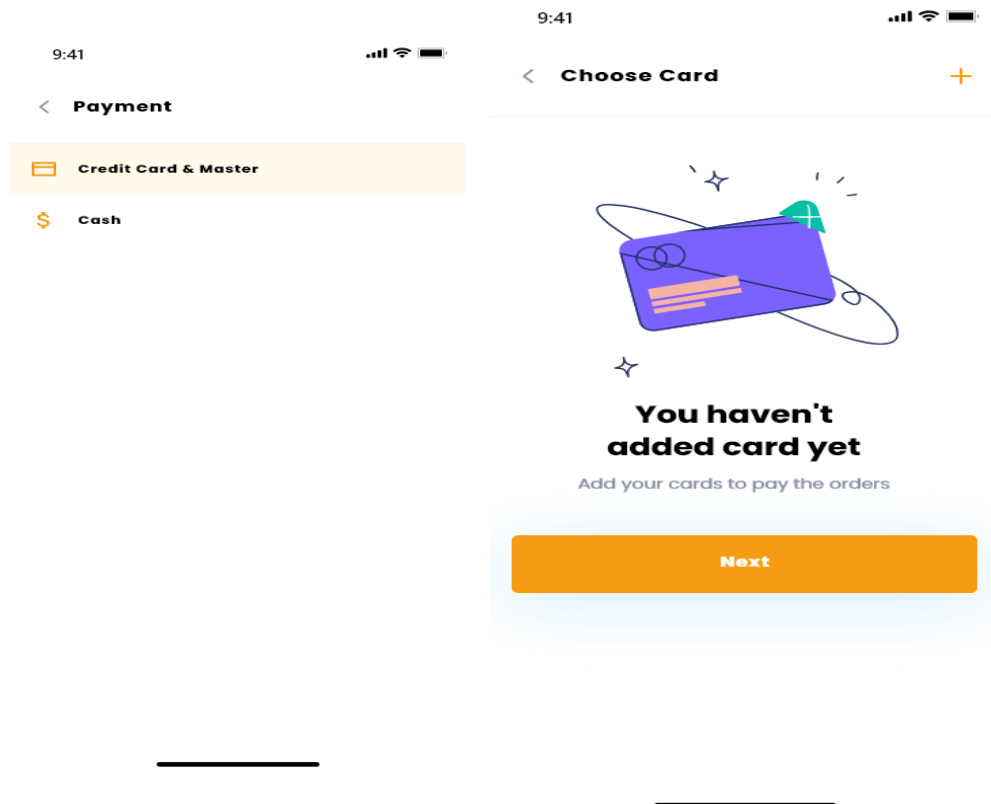


Figure 5.20 Categories Screens Design.

Payment screen and you choose the appropriate payment method for you and then the order details page appears when you choose payment upon receipt, or the visa page and fill in the required data.



Favorites screen contains the products that you like. The product detail screen contains all the product information.

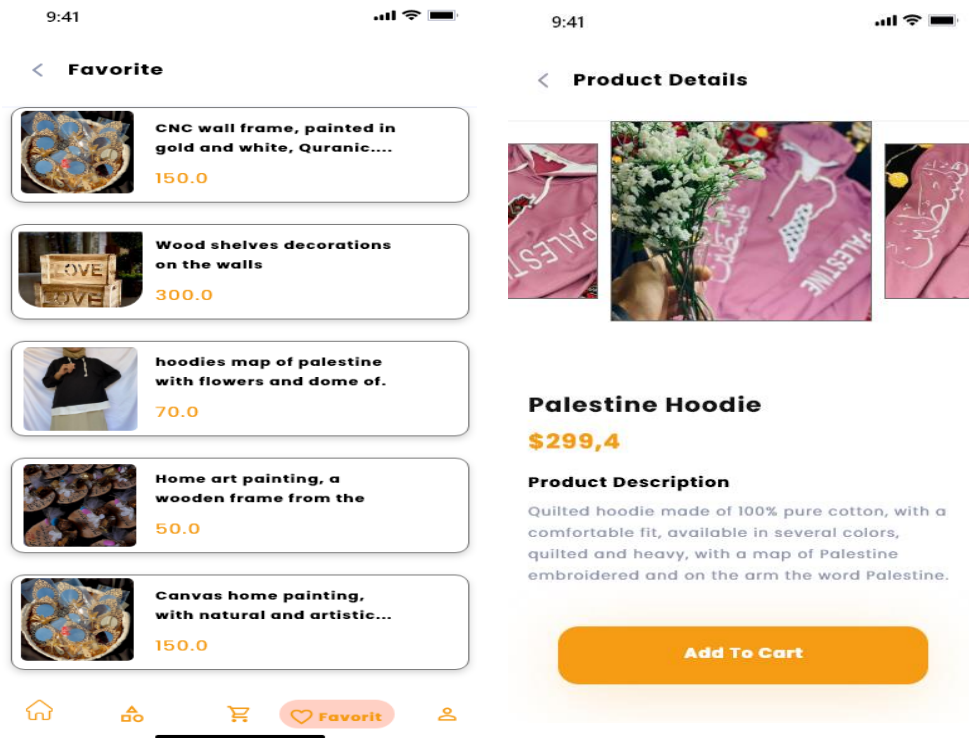
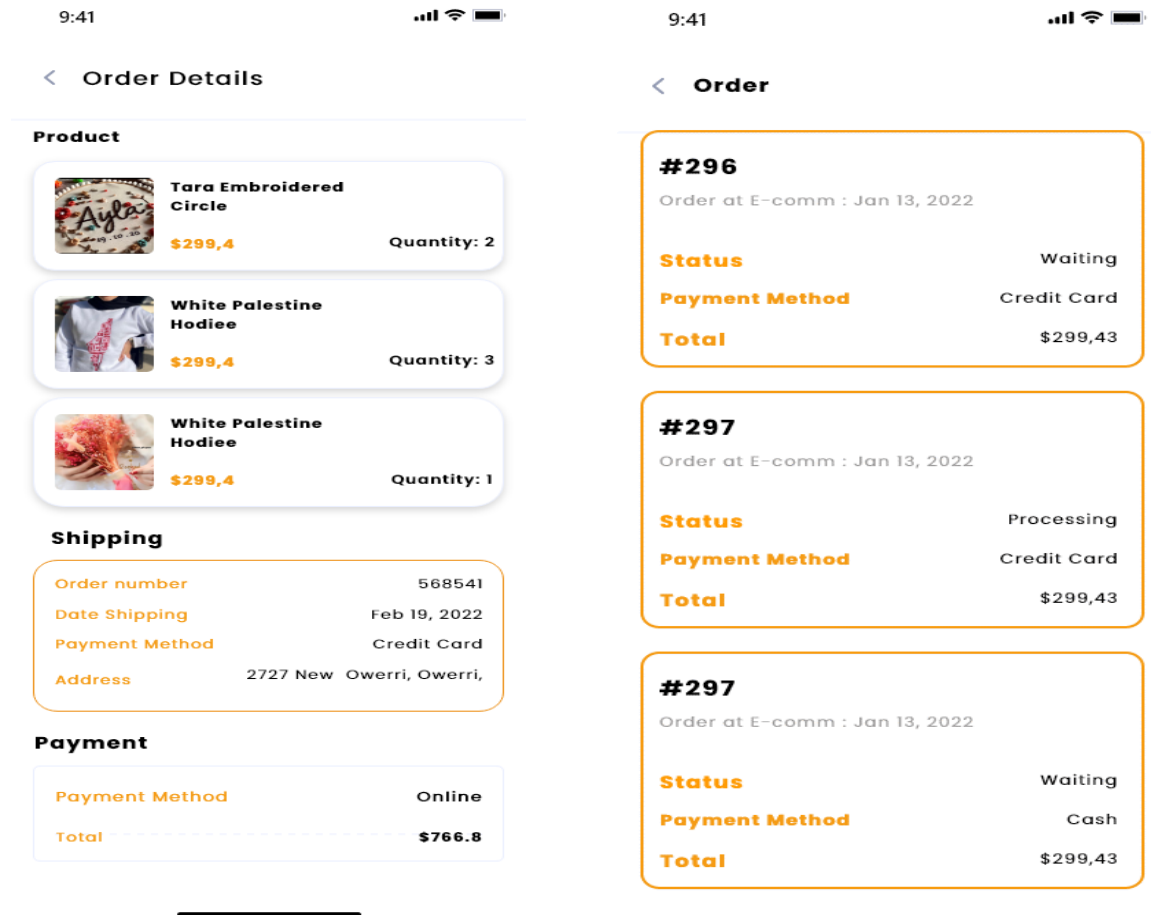


Figure 5.22 Favorite and Proudest Details Screens Design.

Screens for the order process, order details page, order completion and success



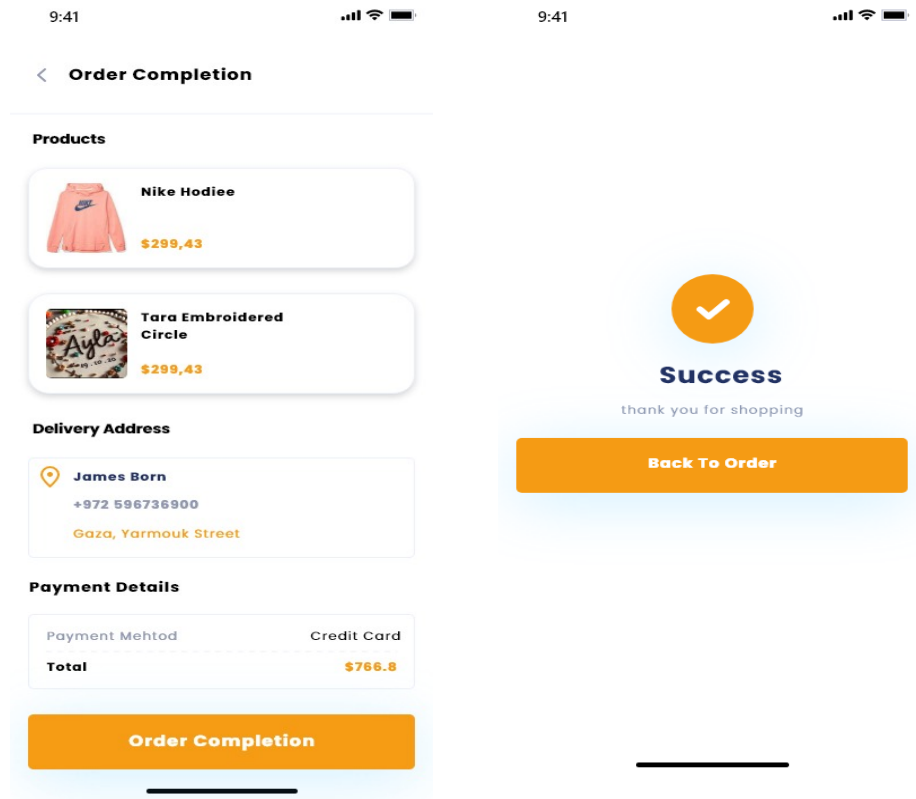
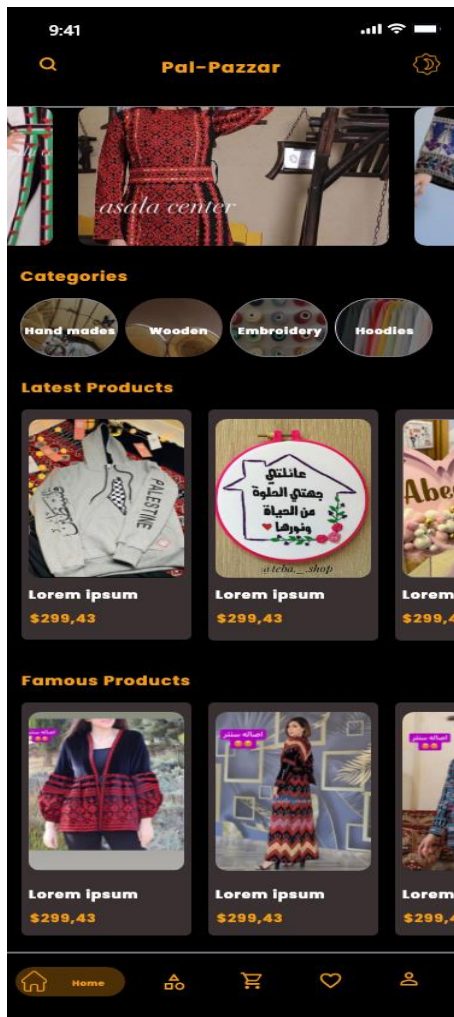


Figure 5.23 Order Screens Design.

Example of Dark Mode pages supported by the application.



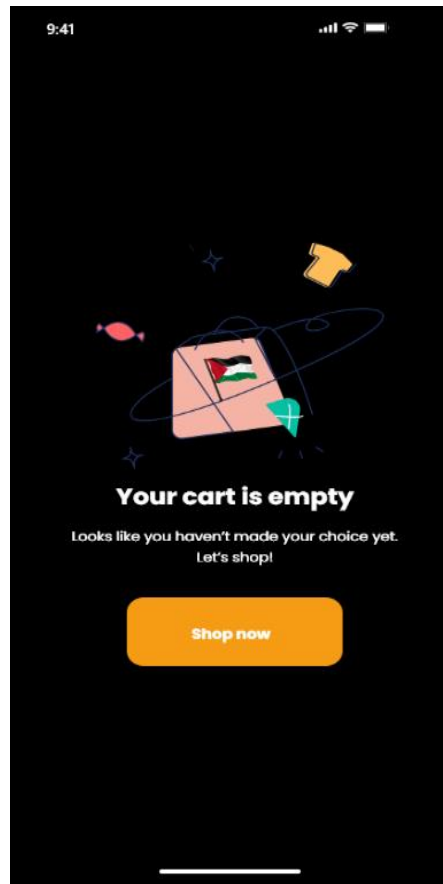


Figure 5.24 App Mood Screens Design.

Chapter 6

Implementation

6.1 Introduction

In this section, we explain the most important phase in the software development process, which is implementation. In this phase, we create the physical source code of the application being built, and we code the application on the basis of the collected requirements and the developed project documentation, then we integrate parts of the source code which we developed into an executable App.

The following section explain the framework that use dart as programming language

6.2 Flutter Framework

Flutter is a cross-platform software development framework that was presented by Google in 2015 and release in May of 2017. Nowadays flutter has steadily grown and provided possibilities not only for IOS and Android mobile development but also for web and desktop applications as well.

6.3 Code Structure

6.3.1 MVC Design Pattern

The Model-View-Controller (MVC) is an architectural pattern that separates an application into three main logical components: the model, the view, and the controller. Each of these components is built to handle specific development aspects of an application.

The MVC component

- **Model**

Models are data classes that you create according to your needs, and with the help of the controller, you can manipulate the model's data.

- **View**

Views are all the Widgets and Pages within the Flutter Application

- **Controller**

The controller layer is high-level functions that perform a specific type of task. They typically fetch from services and inject them into models and that way they control and update app state.

6.2.2 Singleton Design Pattern

Is a pattern used in object-oriented programming which ensures that a class has only one instance and also provides a global point of access to it. Sometimes it's important for a class to have exactly one instance. For example, when only want one instance of a class that represents the local storage.

6.4 Development Tools

In this section, we introduce the tools and technologies we used along the life cycle of our project.

- **GetX State Management:** state management in general, is the strategic approach to manage all the interactions that a user performs on an application and then reflect those changes to UI [17]. GetX is more than just a state management library. It is, in fact, a small flutter framework capable of handling route management and dependency injection in flutter applications.

We have used it because of the many advantages it provides, including:

- **Performance:** Focused on minimum consumption of memory and resources by it doesn't keep the controller in memory by default if it is

not in use, unlike most other platforms, the controller must release once it is not in use to optimize the memory usage.

- **Productivity:** Intuitive and efficient tool combined with simplicity and straightforward syntax that ultimately saves development time [18].
 - **Organization:** Decoupling of view, business logic, dependency injection and navigation makes it better than any other approach to organize the code. GetX manages its own dependencies so we don't require it to inject our own controller, model or blocs classes in the widget tree. Also, do not need context to navigate between routes, nor need stateful widgets [18] [19].
- **RESTful API:** Is an architectural style for an application program interface (API) that uses HTTP requests to access and use data. That data can be used to GET, PUT, POST and DELETE data types, which refers to the reading, updating, creating and deleting of operations concerning resources. An API for a website is code that allows two software programs to communicate with each other. The API spells out the proper way for a developer to write a program requesting services from an operating system or other application [20].
- **JavaScript Object Notation (JSON):** Is a text-based, human-readable data interchange format used for representing simple data structures and objects in Web browser-based code and used for primarily to transmit data between a server and web application [21].
- **Flutter_screenutil (Support Multiple Screen Sizes):** The rapidly wide usage of different mobile devices led to make a problem in the design of

mobile applications, and their suitability for different mobile screens in terms of size and resolution. Hence the need emerged to adopt other methods for designing mobile applications capable of reformulating themselves to suit the multifarious features of diverse mobile, flutter screen util package solved this problem and allow to have full control of every single pixel displayed on the screen.

- **Http Package:** The package contains a set of classes and functions that allow dealing with API calls.
- **Cached Network Image:** To reduce the web server load, and make a quick response from the App side, this package support and save the requests for the images in the cache memory and retrieve them from the cache when called many times.
- **Navigation and Routing:** Are very important for any application. Flutter provides two types of APIs for navigation: imperative and declarative, and we use the imperative routing mechanism in our App, which include **MaterialPageRoute** mechanism that based on the stack-based history of routes, but if there is a need to navigate to the same screen in many parts of the app, this approach is not beneficial because it results in **code duplication**. The solution is by defining the **Named Routes** for navigation [24].
- **Page View:** is a flutter widget that creates scrollable pages on the screen and allows the user to transition between them. We use it in onboarding screens.
- **Shared Preferences:** Flutter package to save a small value that allows you to reference later sometime when a user launches the application. Shared Preferences Supported data types are int, double, bool, String, and List<String> [23].

- **SQLite Database Storage:** Is a relational database management system contained in a C library and it is not a client-server database engine [25]. It is persistent data store that preserves the data even after the application is closed.
- **GetX Themes:** Switching between light and dark themes is made simple with GetX.
- **GetX Storage:** GetX also provides fast and extra light synchronous key-value memory backup of data entirely written in Dart that easily integrates with the GetX core package.
- **Localization:** The approach is to use a reference key in a translation dictionary instead of hard-coded strings. The next time the application displays some text, it looks up the key in the dictionary for the selected user language and retrieves the content accordingly.
- **Marital Design:** Is a visual language that synthesizes the classic principles of good design with the innovation of technology and science.

6.5 Assistant Programs for Writing

- **Gleek Digram Maker:** Is text-to-diagram tool for developers and software architects, used to create UML class, sequence, entity-relationship diagrams [22].
- **Quetext (Plagiarism Checker):** Is extremely effective for writers who need to verify their content's originality. Not only does the plagiarism checker quickly and accurately identify any and all instances of plagiarized writing, but it also generates citations where necessary, right on the spot[26].
- **Flowchart Maker (Digrams.net):** Is free online diagram software used as a flowchart maker, network diagram software, create UML online, as an

ER diagram tool, design database schema, create use cases, as a circuit diagram maker, and more[27].

- **Office Timeline:** Office Timeline is a Microsoft PowerPoint plug-in that allows you to create professional looking timelines and Gantt charts directly inside PowerPoint. It's very simple to use because the application does all the chart creating. All you have to do is enter the milestones and tasks using the new timeline wizard [28].
- **Adobe XD:** Is a desktop program made for designing user interfaces and wireframes for a variety of applications [29].
- **Adobe Photoshop CC Adobe Photoshop:** Is a raster graphics editor developed and published by Adobe Systems for Windows and operating system (OS) [30].
- **Microsoft Office Word 2019:** Is a word processor developed by Microsoft, it offers enhanced features to create professional-quality documents, easier ways to work together with people, and almost anywhere access to your files.

Chapter 7

Testing

Testing is performed to ensure that the programs are executed correctly and conforms to the requirements specified.

Testing in Flutter can be split into three categories as explained in section 7.1:

7.1 Testing Categories

7.1.1 Unit Testing

Unit testing is the easiest method to test an application. It is based on ensuring the correctness of a piece of code (a function, in general) a method of a class. But, it does not reflect the real environment and subsequently, is the least option to find the bugs.

7.1.2 Widget Testing

Widget testing is otherwise called component testing. As its name proposes, it is utilized for testing a single widget, and the objective of this test is to check whether the widget works and looks true to form.

7.1.3 Integration Testing

Integration testing involves both unit testing and widget testing along with external component of the application like database, web service, it simulates the real environment to find nearly all bugs, but it is the most complicated process.

7.2 Test Cases:

Table 7.1 Test Cases Table

ID	Module name	Test steps	Expected data	Actual data	Status
TC01	Register	1. Open the App. 2. Click on Register. 3. Fill the required data. 4. Click on Register button.	Message with verification code and navigate to verify account interface	As expected	Pass
TC02	Sign In	1. Click on sign in. 2. Write phone number and password. 3. Click on sign in button.	Message shows success sign in and navigate to home interface	As expected	Pass
TC03	Forget Password	1. Click on forget password button. 2. Write phone number. 3. Click on send button. 4. The verification code will sent.	Message with verification code and navigate to reset password interface	As expected	Pass

		5. Navigate to reset password interface.			
TC04	Update Profile	1. Click on profile in bottom navigation. 2. Click on update profile. 3. Change the user data. 4. Click on check icon.	Message shows that the data was updated successfully	As expected	Pass
TC05	Search	1. Click on search icon. 2. Search for any product you want.	Show what you searched for.	As expected	Pass

TC06	Change Password	1.click on profile in bottom navigation 2. click on settings 3.click on change password 4. fill in the password, new password, and confirm password 5. click on save button	Message shows that the password was updated successfully	As expected	Pass
TC07	Add to cart	1.choose any product 2.on product details click on add to cart button	Message shows that the product added successfully	As expected	Pass
TC08	Add Address	1. Click on profile in bottom navigation. 2.Click on address 3. Fill the required data. 4.Click on add address button	Message shows that the address created successfully. Show the address in addresses interface.	As expected	Pass

TC09	Add Payment Method	1. Click on payment button. 2. Click on add icon 3. Fill the required data for visa card 4. Click on add button	Message shows that card created successfully and navigate to payment cards interface	As expected	Pass
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7.3 Testing and Results

In this section, we will show the performance of the App in time and memory.

Performance is a crucially important aspect of any modern mobile app. Skipped frames hurt the app's usability and leave a bad impression on users.

The App should maintain 60 FPS (frames per second) most of the time, or in other words, change frames every 16.66 milliseconds.

7.3.1 Performance Overlay

One way to conduct Flutter performance tests is to use the Performance widget. The overlay widget displays two graphs on top of the app. The top 'GPU' graph shows the raster thread performance, in other words, communication between the app's layer tree and the device's GPU. Figure 7.1 shows how CPU resources are utilized and UI thread.

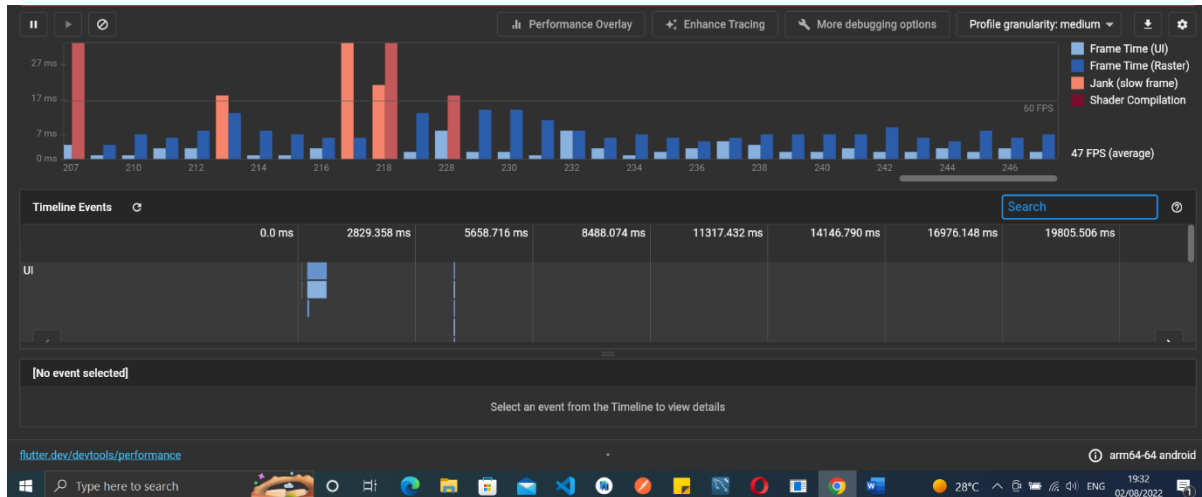


Figure 7.1 CPU Frame Chart

7.3.2 Memory Performance

The stability problems caused by the rising memory level seriously affect the app user experience, so developers should pay close attention to the memory performance of flutter. Idle is also facing the problem of increasing crash rate caused by out of memory. In this section, we will explain how analyze the memory leakage while using the App.

Figure 7.2 shows how active object allocate memory space, each object has it own independent heap and can't share memory with each other, so it can fast allocation without lock.

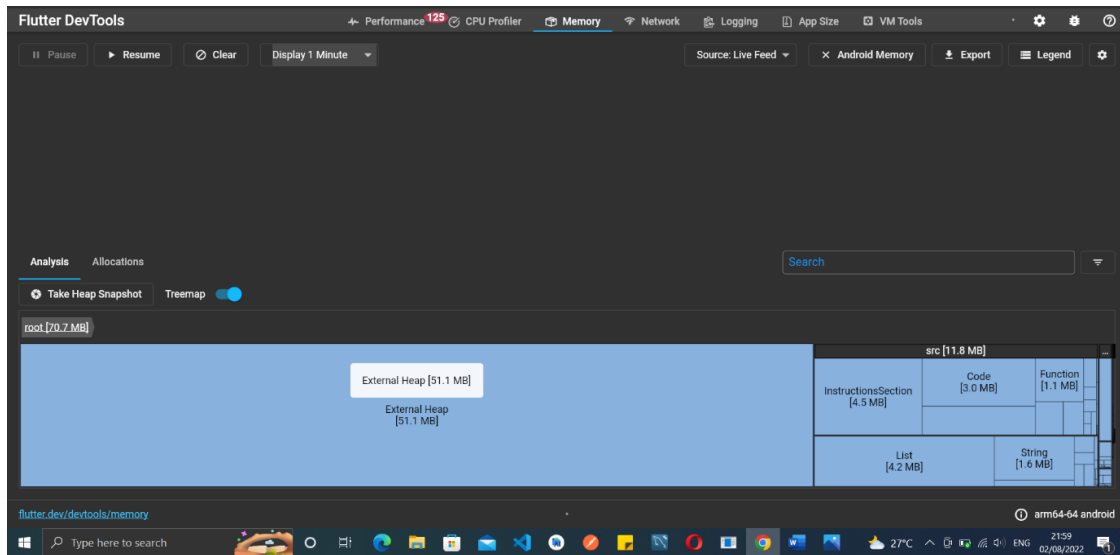


Figure 7.2 Memory allocation

Chapter 8

Conclusion & Future Work

8.1 Conclusion

For now, we present the Pal-Pazzar as the first release and we will continue to develop it in the near future.

This app we developed for Android, and IOS platforms and using Flutter.

This application is integrated with API, APIs provide a secure and standardized way for applications to work with each other and deliver the information or functionality requested without user intervention.

The App has been successfully developed and validated and it fulfils most of the primary functions. The initial schedule was planned at the beginning of the project and after system analysis. This project fulfils 90% of the required jobs successfully.

8.2 Future Work

Innovation and development don't end in the world of mobile applications, so we will not stop continuing to add new features, improvements, and facilities for users until the application becomes the best image we expect it.

On future we will work on the following:

1. We plan to add other features to the App like notifications, track orders, and support more than two languages.
2. We will add more options in payment methods such as: Jawwal Pay.
3. Collaborate with different business owners to display their products in Pal-Pazzar.
4. Making the application not limited to local delivery and making it support ordering and delivery from abroad.

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